

What if the buses are better?

UTILISING REAL-TIME BUS LOCATION DATA TO UNDERSTAND ACTUAL BUS SERVICE PERFORMANCE AND FACILITATE TRANSPORT APPRAISALS

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ABSTRACT

Since 2024, local authorities in England are granted unprecedented powers over local transport matters. This makes them well positioned to tap on the growing demand for access to the city centres post-pandemic. However, local authorities may not have the tools to make the best choices since they do not have a comprehensive picture of how their existing network influence accessibility, and transport appraisal has traditionally relied on scheduled timetables instead of actual service delivery.

This paper addresses that gap by relying on BODS GTFS-RT feeds to build retrospective GTFS datasets to understand actual bus service performance and speculate between hypothetical scenarios across three English cities with different governance models – Bristol and Leeds (Enhanced Partnership) as well as Manchester (franchised).

Bus service performance is investigated using Hansen-style potential accessibility metric, which is used to measure actual accessibility to the city centre and how it deviates due to travel time inaccuracy (TTI) and travel time variability (TTV). Spatial error regression finds that road network characteristics, especially bus lane provision, are the strongest predictors of accessibility and variability outcomes across all three cities, outweighing bus-network-specific factors. Governance model matters less than expected, with only Bristol showing significantly worse TTI and TTV impacts on potential accessibility than Manchester while Leeds is statistically indistinguishable from it.

Hypothetical scenario modelling generally shows that transit-priority infrastructure offers greater accessibility gains than frequency increases. However, since the former is likely inflated because such interventions have been taken in all three cities, Bristol and Leeds could try switching to franchising to increase bus frequencies. Meanwhile, Manchester may be better off focusing on encouraging multimodal journeys since it has likely exhausted much of the benefits from increased frequencies having switched to franchising for several years.

These findings give local authorities an evidence base for weighing the different options that they now have to improve accessibility via public transport.

DECLARATION

I, Muhammad Khalis Bin Samion, hereby declare that this dissertation is all my own original work and that all sources have been acknowledged. It is 9694 words in length.



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Muhammad Khalis Bin Samion

15 August 2026

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LIST OF ACRONYMS AND ABBREVIATIONS

AADF	Average Annual Daily Flows
BODS	Bus Open Data Service
EP	Enhanced Partnership
GMCA	Greater Manchester Combined Authority
GTFS	Google Transit Feed Specification, later General Transit Feed Specification
GTFS-RT	GTFS Realtime
LSOA	Lower-layer Super Output Area
LTA	Local Transport Authority
MSOA	Middle-layer Super Output Area
MTUP	Modifiable Temporal Unit Problem
OD	Origin-Destination
OSM	OpenStreetMap
TTI	Travel Time Inaccuracy
TTV	Travel Time Variability
TTWA	Travel-To-Work Area
WECA	West of England Combined Authority
WYCA	West Yorkshire Combined Authority
WZ	Workplace Zone

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SECTION 1: INTRODUCTION

Miller (2018) maintains that the role of a transport system is to facilitate economic activities and/or social reproduction of life by providing means for people to physically access one another or businesses and institutions, as well as vice versa. That is why transport connectivity, or its lack thereof, is often cited as one of the key reasons that are holding the British economy back, and therefore one of the strategies that should be utilised to unlock the economic potential of our cities in the turbulent post-pandemic and post-Brexit landscape (Johnson and Rollison, 2025; Isaac et al., 2026). For England, improving transport connectivity would mean a greater focus on buses over rail or metro/tram. This is because the bus remains the most-used public transport mode in England post-pandemic, with an average of 28 bus trips per person in 2024 vis-à-vis 21 trips per person by surface rail for non-London residents (Department for Transport, 2025).

Hybrid working in the post-pandemic has not diminished the relevance of city centres as the focal point of local economic activity. National Travel Survey data between 2019 and 2022 data show that commuting frequency has dropped but commuting distances have increased for residents in large cities, suggesting that urbanites are more willing to travel to city centres so long as they don't have to travel every day (Bond-Smith and McCann, 2025). Spending data between 2022 and 2024 further prove that hybrid working arrangements have not killed the city centre (Swinney and Selby, 2025). At the same time, it is likely that the city centre would rebound in relevance again due to the growth of the 'new economy' – a collection of knowledge-intensive sectors that either develop new emerging technologies or seek to apply them in other sectors such as agriculture and manufacturing. Policy research has shown that in the UK, such firms are disproportionately clustered in city centres, likely because of the accessibility to skilled workers (Rodrigues, Vera and Swinney, 2022). On top of that, other professional sectors are also pushing for their employees to return to office as per pre-pandemic arrangements (Chung and Yuan, 2025; The Economist, 2025), suggesting that the city centre remains an important destination to which transport networks should provide good access for all.

Meanwhile, since 2024, several legislations have been passed with the effect of significantly empowering local authorities in England (Johnson and Rollison, 2025). The Bus Services Act 2025 allows *local transport authorities* (LTAs) to take control of bus service provisions via franchising or directly operating bus services. The English Devolution and Community Empowerment Act 2026

further delegates powers on transport infrastructure planning to ‘strategic authorities’ – the overall local authorities of which LTAs are a part. Meanwhile, the Railways Bill (2026) is expected to give LTAs more influence because they must be consulted by the Great British Railways pertaining to rail service provisions in their areas.

Hence, LTAs now have unprecedented powers to improve local transport, especially their bus networks, and maximise economic benefits by enhancing accessibility to the city centres. However, LTAs may not have the right tools to help them choose the best option that they now have to best achieve their goals. This is because most LTAs may not fully understand how accessibility has been impacted by their current local networks due to the lack of available real-time location data and the lack of literature that uses it. Furthermore, evaluating accessibility and appraising hypothetical improvements have traditionally relied on the baseline conditions being the scheduled, static timetable data instead of the actual service delivered to commuters.

This is where this paper aims to make its two-fold contribution, using the public transport networks. In Bristol, Leeds and Manchester as its empirical setting. Firstly, this paper seeks to explore factors that influence actual bus service performance across different bus governance contexts. Specifically, three performance aspects will be explored – the actual potential accessibility to city centre during morning peak hours, as well as deviations in potential accessibility caused by *travel time inaccuracy* (TTI) and *travel time variability* (TTV) respectively. Secondly, this paper explores various improvement scenarios to the transport system, especially buses, and speculate on which measures that can offer the best utilitarian improvements in accessibility to the city centre.

This paper will be organised as such: ‘Literature Review’ further expands on the different accessibility metrics, elucidates the literature gap in accessibility studies, and defines core terms and concepts that will recur throughout this paper. ‘Geographical Contexts’ introduces the study areas of Bristol, Leeds and Manchester. ‘Methodology and Data’, ‘Results and Discussion’, as well as ‘Limitations and Conclusions’ will subsequently follow to round up the paper.

2.1: Accessibility Metrics in Transport Appraisal

While the definition of accessibility is manifold, it can be summarised as “the extent to which land-use and transport systems enable (groups of) individuals to reach activities or destinations by means of a (combination of) transport mode(s)” (Geurs and van Wee, 2004). In Guers’s (2020) review of accessibility metrics within transport appraisal contexts, he identified two of the most common metrics used – the **cumulative opportunity** and **potential accessibility** metrics.

Cumulative opportunity is essentially a simple absolute summation of all opportunities within a time threshold (Guers, 2020). The direct predecessor of this paper, which is the Centre for Cities report on public transport integration in large UK cities except London, uses this metric to compare impacts of different hypothetical improvement scenarios on accessibility to the city centre (Johnson and Rollison, 2025). The simplicity of this metric however masks its major shortcoming – the *Modifiable Temporal Unit Problem* (MTUP), where results are affected based on manipulation along temporal dimensions (Pereira, 2019). While it may be convenient to define accessibility based on how many opportunities are reachable within a 30-minute journey, that renders opportunities requiring 35-minute journeys as ‘inaccessible’ when most people would barely register it as an inconvenience. This arbitrariness opens the room for policymakers to adjust the threshold to obtain a more flattering accessibility benefit when promoting a transport intervention to the public.

On the other hand, potential accessibility, as developed by Hansen (1959), can be described as the sum of products between the opportunities at a zone and the cost function for that zone. While it has been originally conceived to measure accessibility from an origin (for example, Verduzco-Torres and McArthur, 2026), it can also be flipped to measure accessibility to a destination (Wu et al., 2019; Soukhov et al., 2025). Unlike the cumulative opportunity metric, potential accessibility sidesteps the MTUP because accessibility is moderated by a parameter β . In turn, this parameter can be calibrated in the same way as one does for spatial interaction modelling.

Klar et al (2023) and Ojha et al (2026) compared both metrics in their respective research and found that potential accessibility has a greater ability to register accessibility changes than cumulative opportunities. Therefore, even if potential accessibility findings are less interpretable (Soukhov et

al., 2025), this paper adopts the position that it is the more robust metric for ex-ante transport appraisals and will use it to achieve this paper's objectives – measuring various aspects of actual bus service performance and comparing between hypothetical improvement scenarios.

2.2: Use of Real-Time Data in Accessibility Studies

Google Transit Feed Specification, later *General Transit Feed Specification* (GTFS), is a static dataset comprising of at least seven relational text files that transport agencies or operators use to share information about their scheduled services, such as which routes do they operate and when do they run them (MobilityData, 2026). There is also a real-time extension called *GTFS Realtime* (GTFS-RT). Unlike its static counterpart, GTFS-RT data is provided in binary files that needs prior decoding and only contain the buses' live locations and the trips that they are serving (MobilityData, no date). Some GTFS-RT feeds would also have information about the next stops to which these buses are heading and their estimated arrival times, but the feeds provided by English bus operators are known to have left this out (thienthanh, 2025; Webb et al., 2025). This means that static GTFS data is favoured over the dynamic GTFS-RT data within academic circles because it is easier to access and interpret.

The path dependency of preferring GTFS data to GTFS-RT is further cemented by the advent of *R5*, a Java-based routing model, and its wrappers in *R* and *Python* languages (*r5r* and *r5py* respectively), all of which requires public transport input to be in the static GTFS format. *R5*'s appeal lies in its ability to produce travel time estimates for multiple origins and destinations quickly (Conway, Byrd and van der Linden, 2017), meaning that it is a useful tool for transportation researchers to measure accessibility. It has been applied for various contexts around the world such as Brussels, Santiago and Mexico City, to name a few (Verbeken, Vanhoyweghen and Ginis, 2025; Marin-Flores, Ferres and Tenkanen, 2026; Verduzco-Torres and McArthur, 2026). In the UK, it has been used to generate nationwide accessibility indicator datasets (Verduzco-Torres and McArthur, 2024), evaluate transport infrastructure investments (Lee and van Dijk, 2026) and explore healthcare accessibility inequality (Chen and Botta, 2025).

However, static GTFS data does not reflect the dynamic nature of public transport operations. In reality, delays could arise at any part of the route for multiple reasons such as long boarding times, stopping at successive traffic signals, or short distances between consecutive stops (El-Geneidy, Horning and Krizek, 2011; Park et al., 2019; Kaewunruen, Sresakoolchai and Sun, 2021). These

delays may cascade down the route, knocking services off their schedules. Meanwhile, static data may also incorporate schedule padding, where extra time is added to accommodate for potential delays such as the ones mentioned earlier (Wessel and Farber, 2019). That means that findings from accessibility studies based on static GTFS data alone are inaccurate because they could either be overestimations or underestimations. It was Wessel, Allen and Farber (2017) who first developed a method to convert GTFS-RT data into retrospective GTFS datasets and then demonstrated its utility by exploring how the divergence of actual performance from scheduled services impacted cumulative accessibility to employment. This paved the way for other researchers to apply and adapt their methods to measure actual accessibility in different contexts (Braga, Loureiro and Pereira, 2023; Javanmard et al., 2025).

In the UK, the viability of real-time data utilisation in accessibility research increased after the introduction of the *Bus Open Data Service* (BODS) in 2021, which mandated all bus operators, among other things, to share their live location data. However, it remains limited because there was no stable archive of BODS GTFS-RT feeds before June 2025 (Open Innovations, 2025).

Nevertheless, this paper is riding on a growing domestic interest to utilise real-time data in accessibility, evidenced by the recognition of how actual service divergence from scheduled could be one of the reasons behind UK cities' lagging productivity vis-à-vis their similarly sized European counterparts (Strange, 2024), and the development of code to convert BODS GTFS-RT feeds into retrospective GTFS dataset along the lines of Wessel, Allen and Farber (Open Innovations, 2024; Chen and Botta, 2026).

2.3: Defining Travel Time Inaccuracy (TTI) and Travel Time Variability (TTV)

With the ability to use real-time data, this paper will evaluate more aspects of bus service performance than just the actual potential accessibility to city centre. It will also aim to establish how actual services deviate from scheduled, and how consistent they are over time.

TTI strictly refers to how “scheduled levels of service might differ from what is delivered to the population” (Braga, Loureiro and Pereira, 2023). That means that actual service may be worse or better than scheduled. TTV, on the other hand, looks at the variability of travel times between an origin and a destination. This variability could manifest between two modes of transport that start their journey at the same time, or between different times of the day, or over different days (Noland and Polak, 2002; Mazloumi, Currie and Rose, 2010). This paper would look at TTV over different

days, since the mode and time-of-day focus has been specified to buses during morning peak hour periods.

While the causes of TTI and TTV may be the same, and TTI could induce TTV, Chen and Botta (2026) are clear that both terms by themselves are ultimately separate issues. This is because a service that consistently runs five minutes late every day is a representation of TTI, not TTV. Meanwhile, a punctual service that has different frequencies depending on the time of day does not have TTI but has TTV.

Research that focuses on TTI or TTV is rarely centred on England – Mazloumi, Currie and Rose (2010) in Australia; El-Geneidy, Horning and Krizek (2011) as well as Liu, Porr and Miller (2023) in the United States; Wessel, Allen and Farber (2017) in Canada; Braga, Loureiro and Pereira (2023) in Brazil. When they do, they either have a small scope (Kaewunruen, Sresakoolchai and Sun (2021) investigated causes of TTI on only one route in Birmingham) or do not focus on employment (Chen and Botta (2025) evaluated TTV-impacted accessibility to healthcare facilities). This strengthens the unique value of this paper’s contribution by focusing on different English cities’ and, by extension, different governance models’ accessibility to city centres.

2.4: Defining Routes, Trips, Patterns, and Frequency

As mentioned in Section 2.2, a static GTFS dataset contains seven relational text files that contains information about public transport services that are being operated. All seven files need to be read together because read together because each file alone contains partial information. To illustrate:

- *routes.txt* only contains information about the routes and the operator(s) managing them
- *trips.txt* only contains information about the trip counts that will happen/has happened for each of those routes
- *stop_times.txt* only contains information about the stop arrival and departure times, and its sequence during a trip, for each of those trips
- *stops.txt* only contains the stop coordinate information

Therefore, an operator can generally advertise a route as a ‘high-frequency service that connects between point A to point B’ but looking across the text files would reveal that there are multiple service patterns – defined as a full set of stops that make up a trip, arranged in sequence – that

have their own frequencies but receive the same route branding. This means the frequency of a route is not consistent across every part.

For example, in Greater Manchester, the Bee Network operates Route 192, which stretches from Manchester to Hazel Grove. However, retrospective GTFS data built from BODS revealed that there are actually 11 service patterns during a typical morning peak hour.

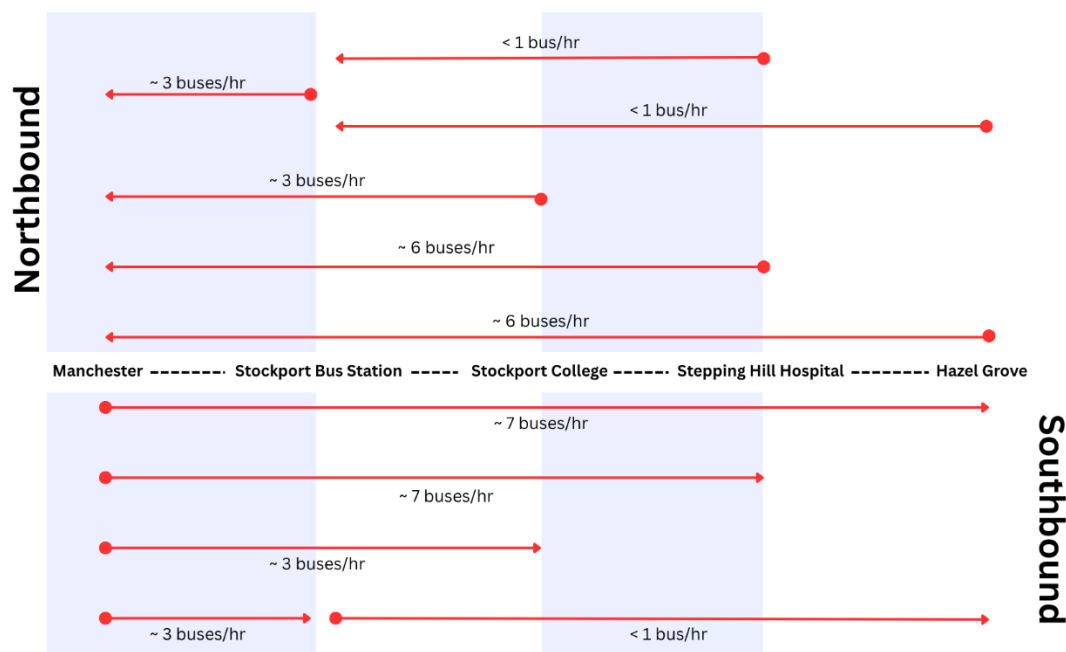


Figure 1: Diagrammatic visualisation of Route 192 service patterns and frequencies in Greater Manchester during typical morning peak hours

Based on Figure 1, in the northbound direction, the last stretch between Stockport Bus Station and Manchester could see up to 18 buses per hour but the first stretch between Hazel Grove and Stepping Hill Hospital only sees 7 buses. This is not a bad thing as it reflects a common practice called ‘short turning’, where buses may run on some parts of a route instead of the full extent so that they can turnaround more quickly to cover specific stretches which may be known to have high ridership demand (Furth, 1987). However, it means that when considering ‘frequency’, this paper does not look at the route’s frequency as a whole and generate one median or one mean statistic. Instead, this paper would consider frequencies of each service pattern.

SECTION 3: GEOGRAPHICAL CONTEXTS

For this paper, three cities will be studied. The first city is Bristol, which is part of the *West of England Combined Authority* (WECA). The second city is Leeds, which is part of the *West Yorkshire Combined Authority* (WYCA). The third city is Manchester, which is part of the *Greater Manchester Combined Authority* (GMCA).

The three cities are selected because they cover the two main ways in which buses are governed outside of Greater London – *Enhanced Partnership* (EP) and franchising. They are differentiated by which of the two stakeholders – bus operators or LTAs – benefits from the power imbalance. In the EP model, which is adopted by WECA and WYCA, the power imbalance favours bus operators over LTAs because bus operators are obliged to uphold a set of standards, such as time and frequency of operations in exchange for LTAs investing in infrastructure that promotes bus ridership (Tyers and Stewart, 2024). Operators, while incurring costs such as fleet maintenance and driver salaries, still get to keep the farebox revenue. LTAs meanwhile do not receive any direct benefits, and their power to make infrastructural changes were limited before the English Devolution and Community Empowerment Act 2026.

In contrast, the franchising model favours LTAs over bus operators because LTAs have full powers over route planning and can dictate to operators on the required service standards (Tyers and Stewart, 2024). LTAs are also the ones who get to keep the farebox revenue, while operators receive a fee for running the services that is determined during the tendering process. As of June 2026, only GMCA has franchised their public bus services.

However, using the political boundaries of GMCA, WECA and WYCA to achieve this paper's objectives would not make sense because it does not reflect the full commuting patterns of its residents – some residents in the periphery areas may work in city centres of neighbouring combined authorities and therefore may not mind a poor accessibility to its own city centre, while other residents from outside the combined authorities' boundaries do work in its city centre and would benefit from improved accessibility to it.

Using Census 2011 *Travel-To-Work Area* (TTWA) boundaries therefore could be an alternative. TTWA is a geographical unit created by Office for National Statistics that approximates self-contained labour markets, where most (approximated at 75%) of the in-boundary residents work within the

boundary and most of the in-boundary workers live within the boundary (Prothero, 2016). The building blocks of TTWAs are *Lower-layer Super Output Areas* (LSOAs). However, measuring potential accessibility requires *Origin-Destination* (OD) data, and current data limitations restrict its granularity to *Middle-layer Super Output Areas* (MSOAs). Since the TTWA and MSOA boundaries do not neatly align, this paper defines the Bristol, Leeds and Manchester study areas as the collection of Census 2021 MSOAs that intersect with the Census 2011 TTWAs respectively (see Figures 2, 3 and 4). The outcome is that the study area will be slightly larger than the Census 2011 TTWAs.

Meanwhile, the city centre boundaries are adapted from Swinney, Selby and Kovacevic (2025). In their research on the performance of high streets across the UK, they defined city centres based on a contiguous collection of Census 2011 *Workplace Zones* (WZs). This paper extends those boundaries to the collection of LSOAs that intersects those WZs to facilitate travel time modelling and OD flow data processing, as will be explained later in Section 4.

3.1: Bristol Study Area

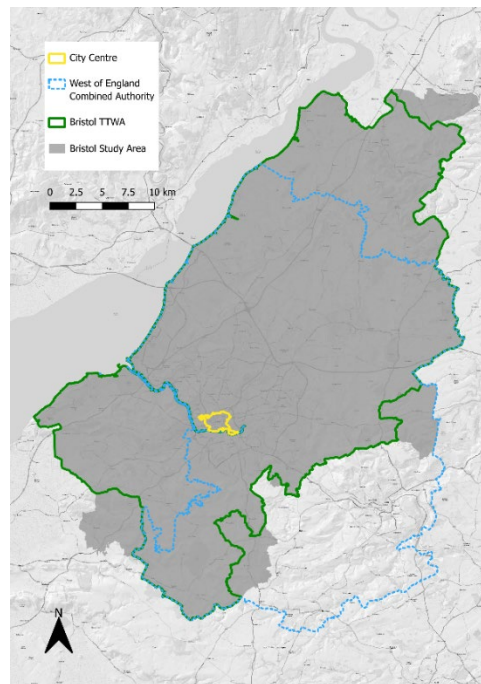


Figure 2: Map of Bristol Study Area and Bristol City Centre vis-a-vis Bristol TTWA and WECA

The Bristol Study Area consists of 102 MSOAs, covering an area of 606.86 km². First Group (officially branded as First Bristol, Bath & the West) is the dominant operator within the area, running 57

routes during the morning peak hours. However, BODS also noted that the following operators have services that are fully within, or enters, the study area during the morning peak hours:

- Eurocoaches – 9 routes
- Faresaver – 3 routes
- Newport Bus – 1 route
- Pulhams Coaches – 1 route
- Stagecoach Group (combining Stagecoach South Wales and Stagecoach West) – 12 routes
- The Big Lemon – 3 routes

Bristol city centre is about 3.64 km², intersecting six MSOAs in the study area.

3.2: Leeds Study Area

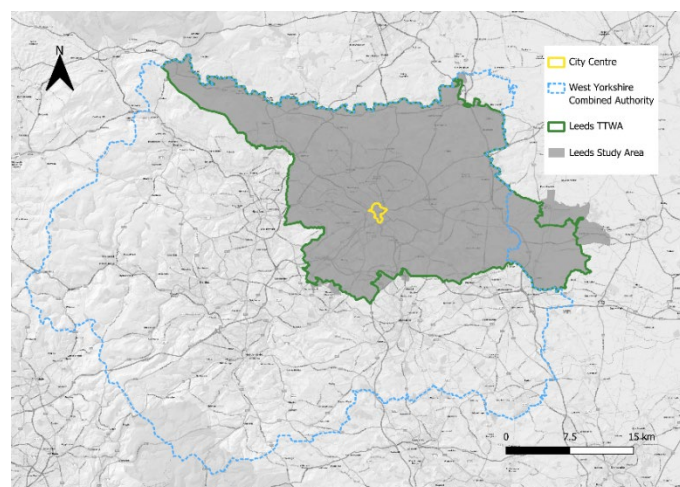


Figure 3: Map of Leeds Study Area and Leeds City Centre vis-a-vis Leeds TTWA and WYCA

The Leeds Study Area consists of 115 MSOAs, covering an area of 679.02 km². Arriva Group and First Group (combining First Leeds, First Bradford and First Halifax) are the two most dominant operators in the area, running 30 and 53 routes respectively. However, BODS also noted that the following operators have services that are fully within, or enters, the study area during the morning peak hours:

- Coastliner – 2 routes
- Connexions Buses – 11 routes
- East Yorkshire – 7 routes

- Flyer – 5 routes
- Squarepeg – 7 routes
- Station Coaches – 1 route
- TLC Travel – 1 route
- The Harrogate Bus Company – 2 routes
- The Keighley Bus Company – 8 routes

Leeds city centre is about 2.15km², intersecting only one MSOA in the study area.

3.3: Manchester

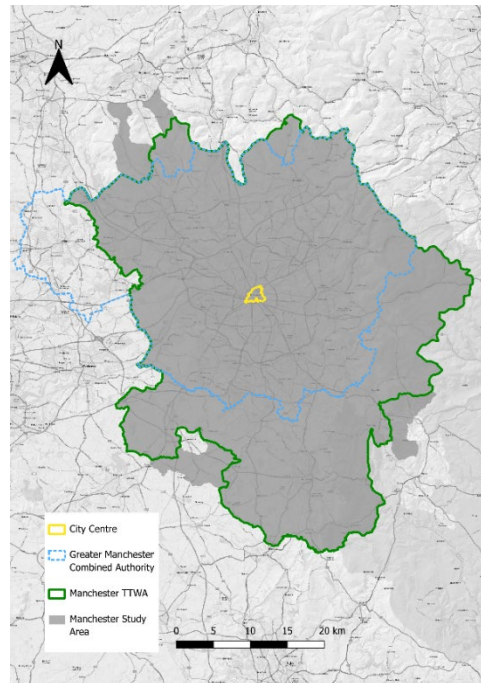


Figure 4: Map of Manchester Study Area and Manchester City Centre vis-a-vis Manchester TTWA and GMCA

The Manchester Study Area consists of 343 MSOAs, covering an area of 1557.79 km². Within the GMCA parts of the study area, morning peak hour bus service is predominantly operated by the Bee Network, the franchised branding for buses in GMCA, with 471 routes. However, BODS also noted that the following operators have services that are fully within, or enters, the study area during the morning peak hours:

- Belle Vue Coaches – 3 routes

- D & G Bus – 10 routes
- First Group (based in Halifax, West Yorkshire) – 1 route
- High Peak – 7 routes
- SightSeeing Manchester – 1 route
- Stagecoach Group (based in Cumbria and North Lancashire) – 2 routes
- Team Pennine – 1 route
- The Blackburn Bus Company – 3 routes
- The Burnley Bus Company – 3 routes
- Warrington's Own Buses – 5 routes

Manchester city centre is about 3.3km², intersecting four MSOAs in the study area.

4.1: Retrospective and Scheduled GTFS Datasets

This paper seeks to contribute by evaluating actual bus performance using real-time data and comparing hypothetical scenarios based on improvements from actual service delivery. Therefore, the primary data source will be Open Innovations' BODS archive, which archives GTFS-RT data for all buses in England since June 2025. However, as mentioned in Section 2, GTFS-RT data is not provided in the same structure as static GTFS data, which is preferred when doing accessibility research using *R5*. As such, this paper will use the open-sourced *rt2gtfs Python* package (Chen and Botta, 2026) to convert GTFS-RT data from BODS into a retrospective static GTFS dataset that reflects the actual bus operation of a particular day.

Specifically, four Wednesdays in June 2026 have been selected as the days for which GTFS-RT data during the morning peak hours (between 0700hrs and 1000hrs) will be extracted and converted into retrospective GTFS. They are June 3, June 10, June 17 and June 24 respectively. This is to reflect the morning commuting patterns at its busiest, when most workers would be the least likely to work from home for the week and schools are still in session.

To facilitate comparisons with scheduled travel times, scheduled GTFS datasets for the four stated dates will also be obtained from Open Innovations' BODS archive, which also archives scheduled GTFS daily. Travel time modelling will also include rail GTFS data, obtained as a CIF file from Network Rail and converted to GTFS using *uk2gtfs R* package (Morgan and O'Neill, 2026), and Metrolink tram GTFS data (only for Manchester and available via BODS archive) to provide a comprehensive travel time landscape for each study area.

4.2: About Travel Time Modelling

This paper shall follow the accessibility research tradition by using *R5* to model travel times between origins and the city centres. Specifically, this paper shall use the Python wrapper, *r5py* (Fink et al., 2022). *r5py* requires the following types of data as inputs:

- Road network data from OpenStreetMap in PBF format. This has been extracted from Geofabrik GmbH (no date) and cropped to each of the three study areas separately.

- Public transport schedules in static GTFS format, to be customised based on scenario to be modelled
- Point coordinates for origin(s) and destination(s)

Typically, *r5py* will iteratively find the best route from the origin point to the destination point based on the parameters that needs to be preset. This paper expands on Johnson and Rollison (2025) to define the parameters of a ‘reasonable’ public transport journey in *r5py* as follows:

- ``departure`` = 0700hrs on each Wednesday
- ``departure_time_window`` = 180 minutes, to cover departures across the entire morning peak hour period
- ``transport_modes`` = transit (which includes bus, rail, and specifically for Manchester, tram) and walk
- ``max_public_transport_rides`` = 1
- ``max_time`` = 180 minutes, to ensure that at least one journey made during morning peak hours ends during morning peak hours

Each travel time matrix produced will therefore feature the median travel time from the origin to the destination for one of the Wednesdays.

This paper shall also follow Verduzco-Torres and McArthur (2024) in setting the origin points as the population-weighted centroids of Census 2021 LSOAs within study areas and the destination points as the population-weighted centroids of Census 2021 LSOAs within its respective city centres. Hence, the MSOA travel time to the centre would be the median of all its-LSOAs-to-city-centre-LSOAs modelled travel times, depending on temporal scope. For example, the ‘overall actual median travel time’ will be the median of LSOA-to-LSOA modelled journeys across all four Wednesdays based on retrospective GTFS datasets but ‘17 June actual median travel time’ will only be the median of LSOA-to-LSOA modelled journeys based only on 17 June’s retrospective GTFS data.

4.3: Calculating Potential Accessibility

As mentioned in Section 2, this paper will measure accessibility to the city centre using Hansen’s (1959) potential accessibility metric. Specifically, this paper will use the inverse version of the equation to measure potential accessibility to the destination instead of potential accessibility

from origin. Therefore, the potential accessibility score, V , for each MSOA will be defined as the product between its own mid-2024 population estimates O_i (Office for National Statistics, 2025a), and the median travel times to the city centre (depending on the scenario) c_{ij} raised to the power of a distance-decay parameter β . The higher the score, the greater is its accessibility to the city centre.

$$V = O_i c_{ij}^{\beta}$$

The parameter β is initially calibrated using Poisson regression in the same way as fitting for spatial interaction modelling (Flowerdew and Aitkin, 1982). This requires a trustworthy OD flow dataset, ruling out the Travel-To-Work dataset from Census 2021. Instead, this paper opts to use the full MSOA-to-MSOA OD flow dataset from Zhong et al (2025). This is because it uses mobile phone location data captured during November 2021, a period with relatively higher activity levels, vis-à-vis Census 2021's self-reported data collected during the tail-end of the third national lockdown. Since the granularity is at the MSOA level but this paper defines city centres as a collection of LSOAs, the OD flows are further manipulated using the ratio of an MSOA's employment count that is within the city centre, derived from 2024 Business Register and Employment Survey data (Office for National Statistics, 2025b) to produce specific MSOA-to-city-centre flow counts.

The equation used in the Poisson regression is a modified version of the unconstrained gravity model equation. Since there is only one destination – the city centre – the destination weight is dropped. Since we are modelling one common parameter β to model flows across different areas, the study area categorical predictor D is added to soak the inter-city differences, with Manchester serving as the reference category.

$$\lambda_{ij} = \exp(\text{int} + \alpha \ln O_i + \beta \ln c_{ij} + \delta_1 D_{\text{Bristol}} + \delta_2 D_{\text{Leeds}})$$

The calibrated β value is -3.336.

4.4: Calculating TTI and TTV

As mentioned in Section 2, TTI looks at how actual service may differ from scheduled service. Therefore, this paper evaluates TTI through TTI rate – a measure of difference of overall scheduled potential accessibility from overall actual potential accessibility, as a rate of overall actual potential accessibility. Overall actual potential accessibility refers to potential accessibility scores derived from overall actual median time while overall scheduled potential accessibility refers to scores derived from overall scheduled median time.

$$TTI = \frac{V_{actual} - V_{scheduled}}{V_{actual}}$$

Since TTI can work in both ways (buses running faster or slower than scheduled), a positive TTI rate would indicate that buses in the area generally run faster than scheduled while a negative TTI rate indicates that buses in the area generally are slower than scheduled.

Meanwhile, TTV looks at the variability in actual travel times from an origin to its destination.

Therefore, this paper evaluates TTV through TTV rate – a measure of difference between the highest and lowest day-based actual potential accessibility (actual potential accessibilities for 3 June, 10 June, 17 June and 24 June respectively), as a rate of overall actual potential accessibility.

$$TTV = \frac{V_{max} - V_{min}}{V_{actual}}$$

A small TTV rate would indicate that buses in the area are more consistent while a large TTV rate indicates that bus performance in the area is more varied. Only these four Wednesdays were selected due to computational complexity of modelling multiple hypothetical scenarios over multiple days (see Section 4.6)

4.5: Regression Modelling on Bus Service Performance

The overall actual potential accessibility, TTI rate and TTV rate will be evaluated for its relationship with the following list of predictors, which can be grouped into the following categories:

- Locational
 - Study area, as categorical predictors
 - Median Euclidean distance from the MSOA to the city centre
- Road Network Characteristics
 - Rate of Minor Roads in the MSOA
 - Rate of 'B' Roads in the MSOA
 - Rate of 'A' Roads in the MSOA
 - Rate of dual-carriageway roads in the MSOA
 - Rate of roads within the Primary Route Network in the MSOA
 - Rate of roads with bus lanes in the MSOA
 - Log-transformed density of junctions and roundabouts in the MSOA, per km²

- Bus Network Characteristics
 - Log-transformed density of bus stops in the MSOA, per km²
 - Log-transformed median trip counts serving the MSOA
- Road Usage
 - Log-transformed median 2025 *Annual Average Daily Flow* (AADF) for the MSOA

Most of the rate predictors and the junction density predictor is derived from the Ordnance Survey Open Roads dataset (Ordnance Survey, 2026). The rate of roads with bus lanes is the only rate predictor in which data is scraped from *OpenStreetMap* (OSM). Since OSM is dependent on crowdsourced information, the actual rates of bus lanes in the MSOA are likely to be higher. Meanwhile, the bus stop density is derived from the National Public Transport Access Nodes data (Department for Transport, no date).

To calculate the median 2025 AADF for each MSOA, the actual 2025 AADF data from Department for Transport (2026) is first interpolated to all nodes across the study area before taking the median value.

The median trip count value is derived from the retrospective GTFS datasets, where for each MSOA, service patterns that serve the area are first identified. Following that, for each pattern, its median number of trips that were operated across all four Wednesday morning peak hours are calculated. The median values for those identified patterns are then added together.

Lastly, median Euclidean distance is derived from the median value of all its-LSOAs-to-city-centre-LSOAs Euclidean distances.

A backward elimination method is adopted, where ordinary least squares (OLS) regression from *statsmodels Python* package (Seabold and Perktold, 2010) is done for all the predictors at first, and the predictor with the lowest significance is dropped one-by-one until all remaining predictors are significant at the 10% level. Following that, using *spreg Python* package (Amaral et al., 2026), OLS diagnostic testing is done on the remaining set of predictors to test for OLS assumptions, especially for heteroskedasticity (Koenker-Bassett test) and spatial autocorrelation (Global Moran's I on row-standardised Queen contiguity weights). Should the spatial autocorrelation be statistically significant, robust Lagrange Multiplier tests are then used to choose between a spatial lag or a spatial error model on the remaining predictors should there be significant spatial autocorrelation.

4.6: About Hypothetical Scenarios

Basing off Johnson and Rollison's (2025) research, this paper will also explore the effects on accessibility in the three study areas from two hypothetical improvements to the public transport network, which are as follows:

- Scenario 1 – Reducing In-Vehicle Travel Times: What if, for service patterns with **moderate** (running 5 – 11 trips throughout morning peak hours, roughly between 2-4 buses per hour) and **high** (running 12 trips or more throughout morning peak hours, 4 buses per hour or more) frequencies, its in-vehicle travel time is reduced by 15% from actual performance? This is to simulate the outcome of introducing transit priority infrastructure such as dedicated bus lanes/roads or transit priority signalling.
- Scenario 2 – Increasing Frequency: What if, for service patterns with **moderate** frequency only, a new trip injected at the midtime point of the current headway? This is to simulate the outcome of doubling the frequencies of certain services.
- Scenario 3 – Compounded: Scenarios 1 and 2 combined.

The scope for both scenarios is not meant equal because it reflects the nature of both interventions. With reference to Figure 1, Bee Network could theoretically be specific and add more trips for the service between Manchester and Stockport Bus Station because its frequency is currently deemed as 'moderate'. However, installing traffic priority signalling along the stretch between Manchester and Stockport Bus Station would not just benefit the moderate-frequency Stockport Bus Station-to-Manchester pattern, but also the other moderate-frequency Stockport College-to-Manchester pattern and the high-frequency Stepping Hill Hospital-to-Manchester and Hazel Grove-to-Manchester patterns.

For each scenario, two sub-scenarios are also generated to consider the limited opportunities available to LTAs when deciding on improvement schemes in the post-COVID environment (Erhardt et al., 2024):

- Sub-Scenario 1 – Deprived MSOA Priority: What if Scenario 1 and/or 2 only apply for service patterns that serve most deprived MSOAs (MSOA Index of Multiple Deprivation Decile Rank 1 or 2, based on Just Knowledge's (2025) aggregation of the original dataset from Ministry of Housing, Communities and Local Government)?

- Sub-Scenario 2 – City Centre-Bound Priority: What if Scenario 1 and/or 2 only apply for service patterns that are going to the city centre?

For each day, each city’s retrospective GTFS stop_times.txt file is edited to produce city-specific hypothetical GTFS datasets for every scenario and sub-scenario (see Figure 5). Specifically, editing for Scenario 1 involves compressing every inter-stop segment duration by 15%. Meanwhile, editing for Scenario 2 requires identifying moderate frequency patterns and injecting one hypothetical trip for those routes at the midpoint between two existing consecutive trips. That hypothetical trip’s duration will be built based on the pattern’s median inter-stop segment durations for the day. It should be noted that the edits made to retrospective Manchester GTFS datasets are only on Bee Network-operated patterns, to reflect its already-franchised nature and therefore incorporating the limits of what GMCA can do.

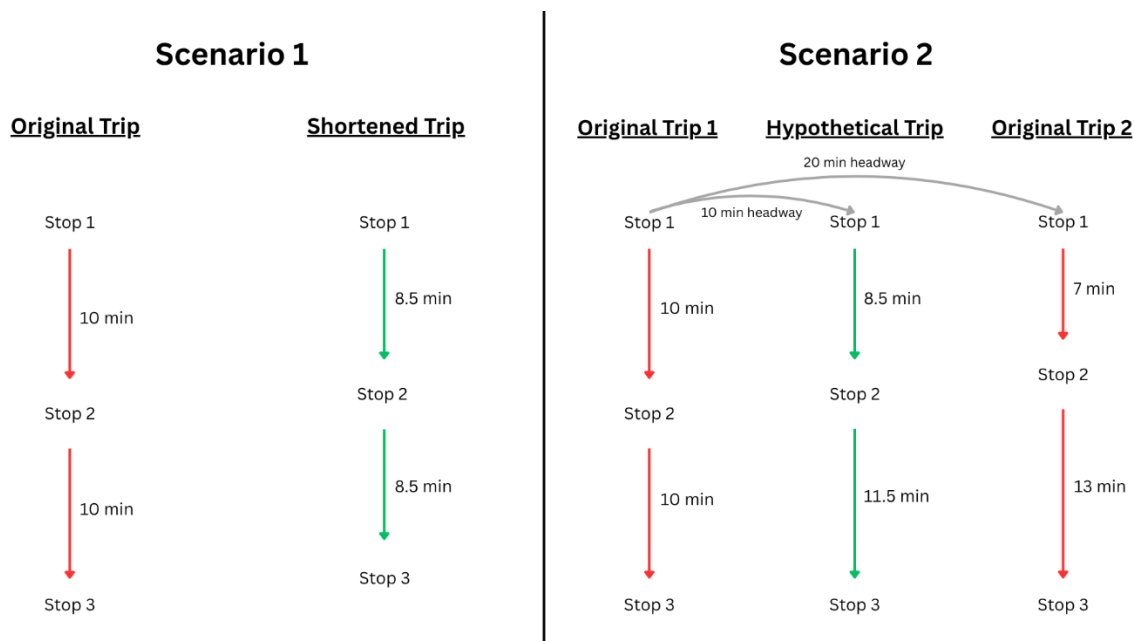


Figure 5: Diagrammatic visualisation of edits to retrospective GTFS stop_times.txt file to reflect Scenario 1 and 2

Every hypothetical scenario/sub-scenario is then modelled twice by *r5py* – once at the same parameters stated in Section 4.2, and another with ``max_public_transport_rides` = 2` – to produce overall ‘hypothetical’ and ‘hypothetical with multimodal’ median travel times, *ceteris paribus*¹. The change in the latter travel time model is to simulate the effects of **encouraging multimodal**

¹ Additionally, the baseline condition based on retrospective GTFS is also modelled with ``max_public_transport_rides` = 2`. This brings the total number of travel time modelling to 63 scenarios [(2 x retrospective, 1 x scheduled, 6 x Scenario 1, 6 x Scenario 2, 6 x Scenario 3) for all three study areas]

journeys via measures like fare capping or syncing of rail and bus schedules to facilitate mode transfers for commuters.

The new median travel times will then be used to derive hypothetical potential accessibility scores. The hypothetical scenarios will be evaluated based on the mean relative percentage improvements in potential accessibility from the baseline condition – the overall actual potential accessibility score. Meanwhile, hypothetical sub-scenarios will be evaluated based on the mean further additions (for multimodal journeys) or subtractions (for deprived MSOA/city centre-bound priority) in percentage points from the main scenarios. It should be noted that the hypothetical potential accessibility scores and relative improvements from baseline should be interpreted as upper-bound counterfactuals.

4.7: Ethical Considerations

As shown in the earlier parts of this section, this data does not rely on any primary data collection from individuals. Most datasets are secondary and open, while Zhong et al's (2025) OD flow data have been anonymised and aggregated to prevent individual-level re-identification risk. The *Python* and *R* packages identified in this section are also used within their scope and cited accordingly.

While the regression and travel time modelling results are presented as evidence that could inform LTAs on capital allocation decision, they should always be weighed alongside actual local constraints and should not be taken as-is. This is especially because this paper sidesteps the question about whose interests should be prioritised in transport appraisals by following the evaluative standard of this paper's direct predecessor (Johnson and Rollison, 2025) and adopting a utilitarian approach, where benefits are maximised, to compare between hypothetical scenarios (Section 4.6). This will be addressed further in Section 6.2.

SECTION 5: RESULTS AND DISCUSSION

5.1: Actual Bus Service Performance

Figures 6-9 show the actual bus service performance for Manchester, Bristol and Leeds respectively.

With reference to Figure 6, there are four Manchester MSOAs and eight Bristol MSOAs which are labelled as “beyond ‘reasonable’ reach”. This means that under actual service patterns, some or all LSOAs in these MSOAs cannot make a journey to their city centres under the parameters specified in Section 4.2. Since potential accessibility values are derived from MSOA median travel times, comparing between scenarios (actual vs scheduled, actual vs hypothetical) for these MSOAs would get distorted because the median travel time for the worse scenarios should have been higher. Hence, to reduce the distortions, these MSOAs would be dropped out from the regressions analysis and hypothetical scenario evaluations.

With reference to Figure 7, the mean and median overall actual potential accessibility score for each MSOA is the highest in Bristol, followed by Leeds and Manchester. However, with reference to Figure 8, the mean and median TTI rate for each MSOA is negative in Bristol while it is barely above 0 in Manchester and Leeds. Furthermore, with reference to Figure 9, the mean and median TTV rate for each MSOA is the highest in Bristol, followed by Leeds and Manchester. This suggests that, after accounting for the local transport networks and road infrastructure, it may be easier for Bristol residents to access its city centre than residents of the other two cities to access theirs, but Bristol residents also suffer more from late and inconsistent bus services than the other two.

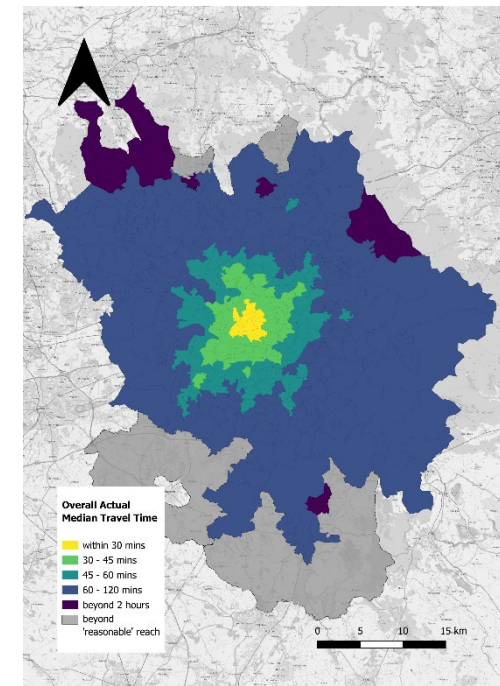
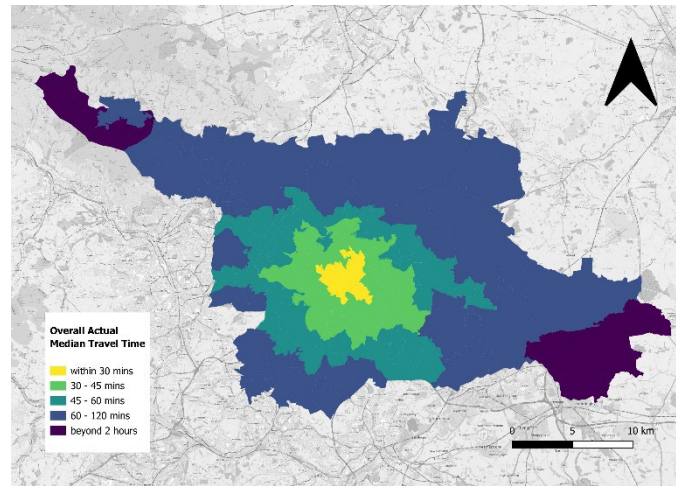
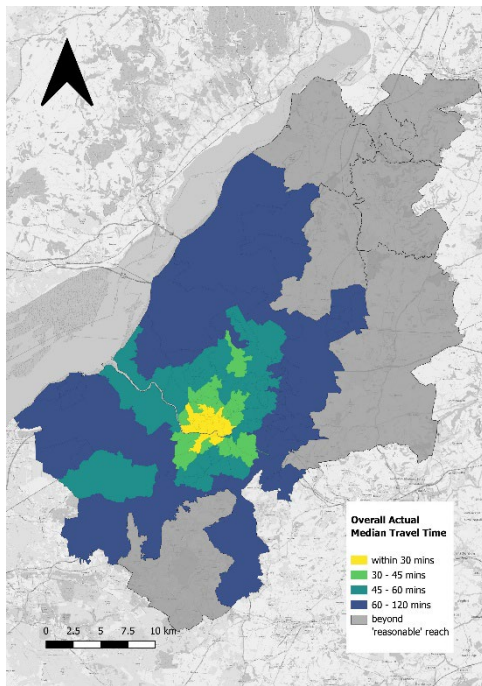


Figure 6: Overall Actual Median Travel Time Results.
From left to right – Bristol, Leeds, Manchester

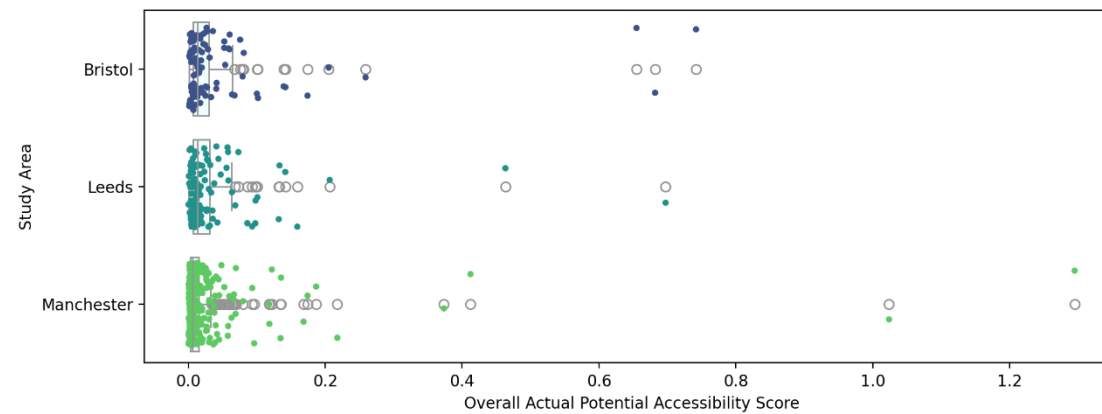
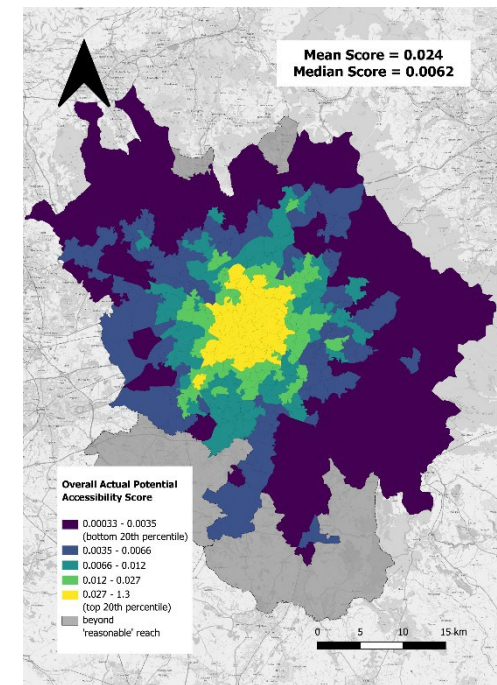
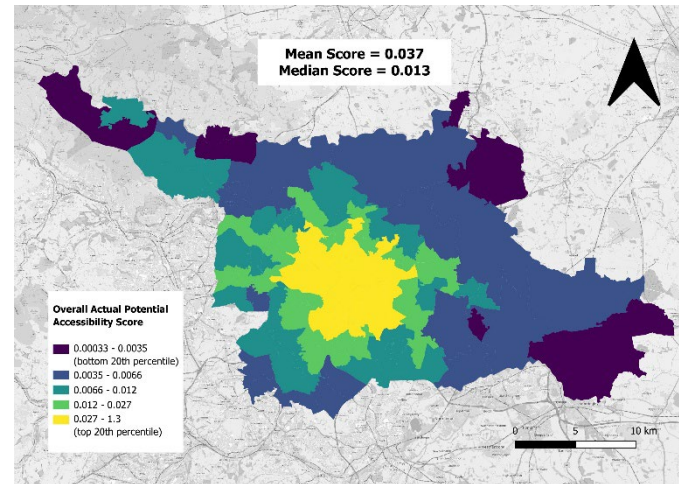
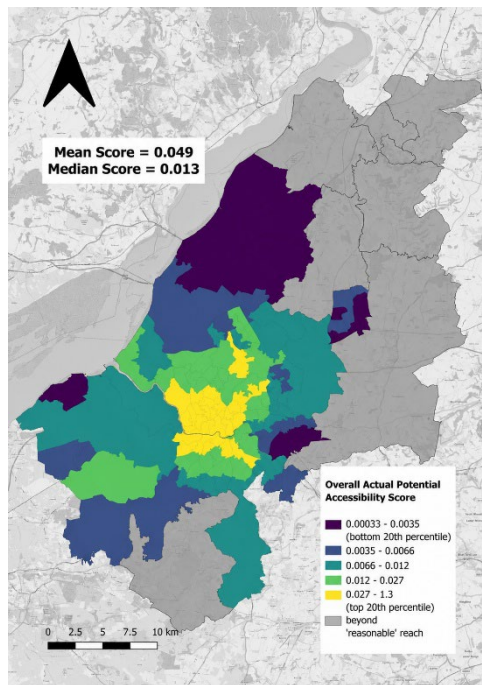


Figure 7: Overall Actual Potential Accessibility Results.
Top, from left to right – Results for Bristol, Leeds, Manchester.
Bottom – Box and strip plot of overall actual potential accessibility score by study area

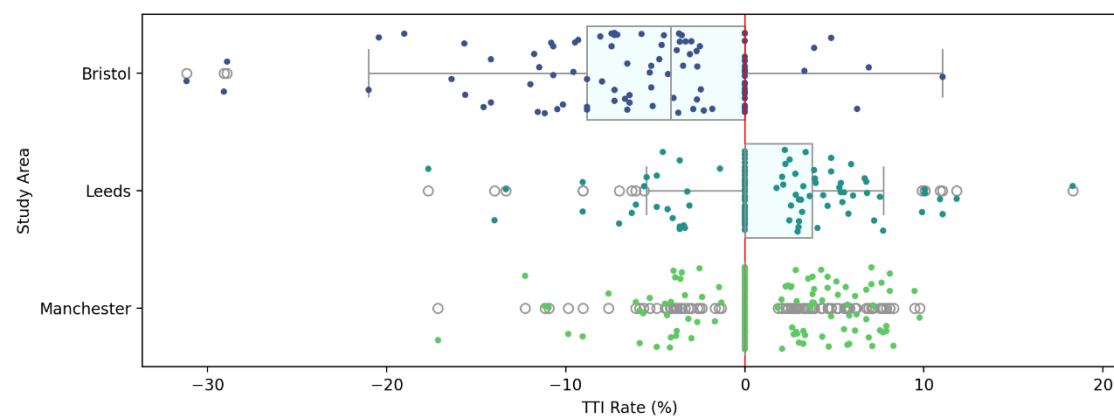
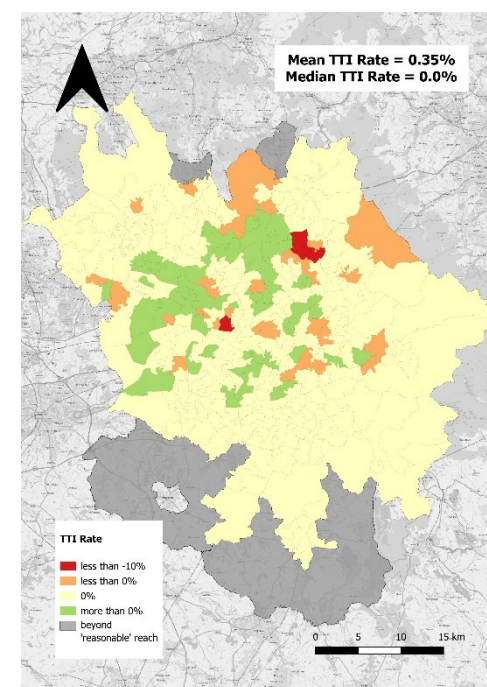
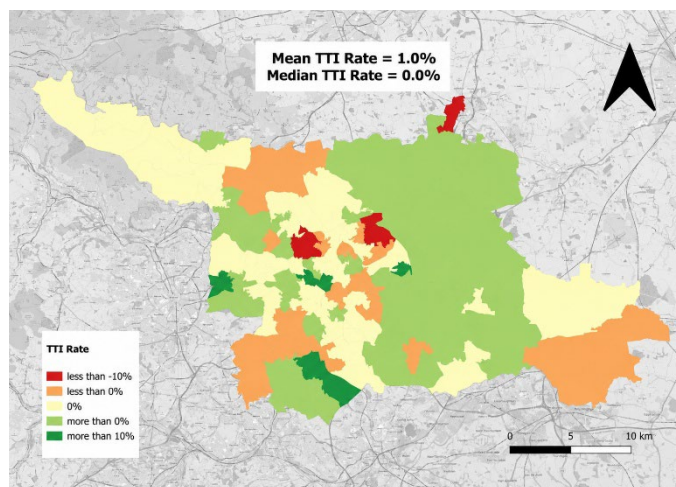
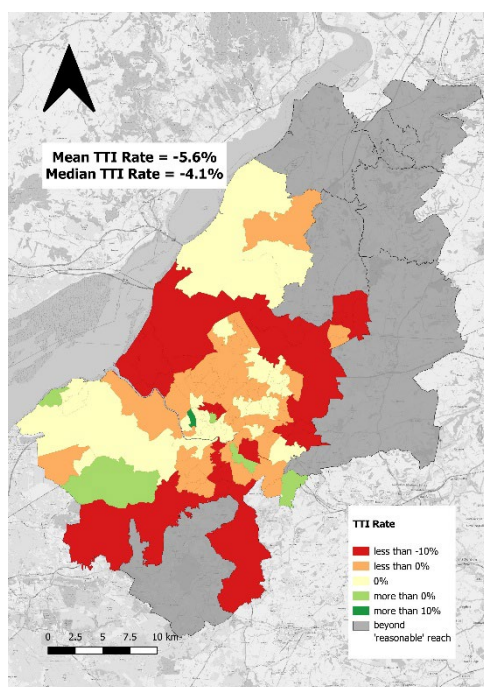


Figure 8: TTI Rate Results.
Top, from left to right – Results for Bristol, Leeds, Manchester.
Bottom – Box and strip plot of TTI rates by study area

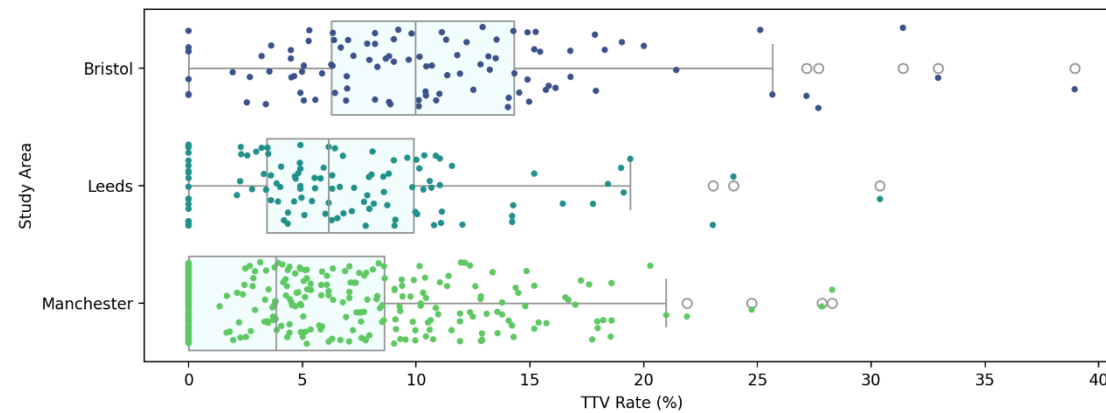
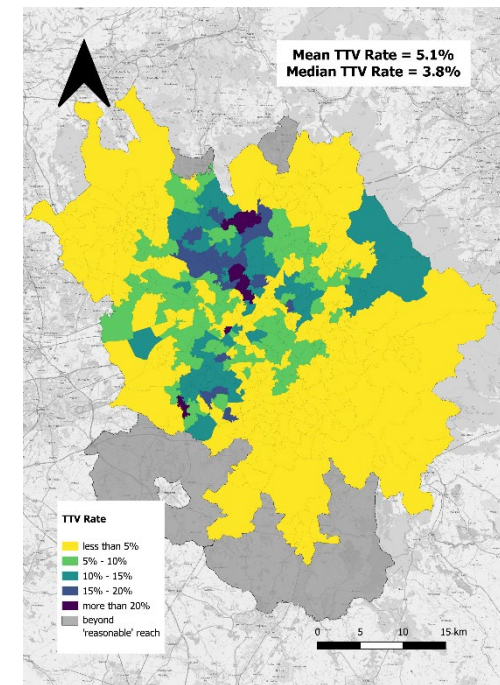
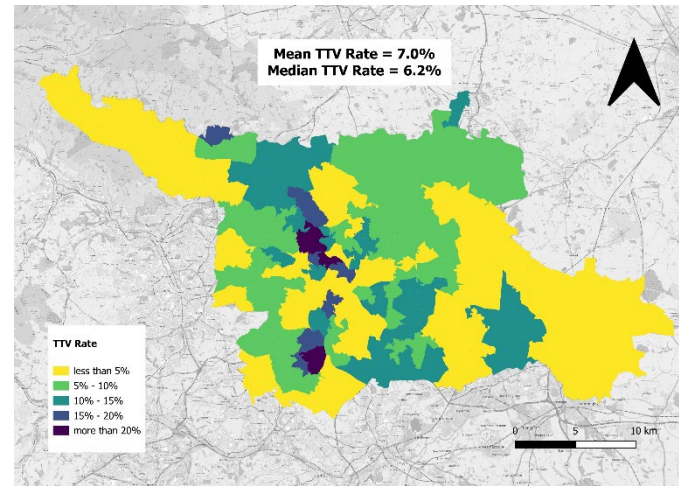
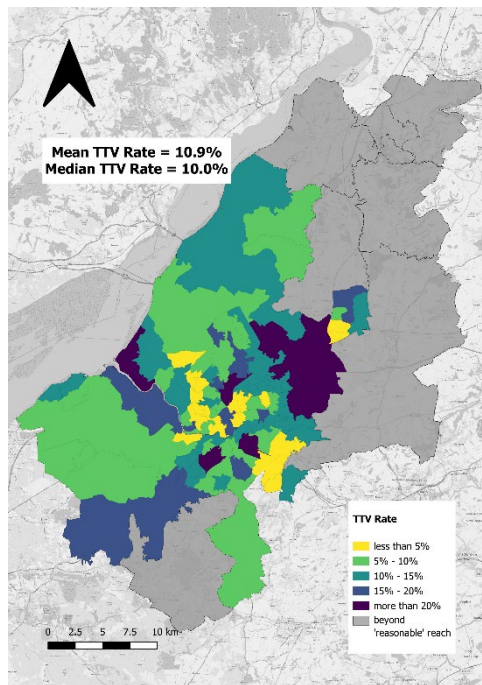


Figure 9: TTV Rate Results.
Top, from left to right – Results for Bristol, Leeds, Manchester.
Bottom – Box and strip plot of TTV rates by study area

5.2: Regression on Bus Service Performance

With reference to Table 1, the OLS regression on log-transformed overall actual potential accessibility found the following factors to have significant relationships at 10% level – percentages of minor roads, ‘B’ roads, ‘A’ roads, primary roads and roads with bus lanes in the MSOA, log-transformed MSOA junction density, bus stop density, median AADF and median trips that serves the MSOA, as well as its median Euclidean distance to the city centre. After OLS diagnostic testing, these set of predictors was found to be heteroskedastic (Koenker-Basset $p < 0.001$), spatial autocorrelation was found to be moderately significant (Moran’s $I = 0.4222$, $p < 0.001$) and robust Lagrange Multiplier tests pointed towards spatial error (228.818, $p < 0.001$, compared to robust spatial lag = 17.035, $p < 0.001$). This justifies the use of a heteroskedasticity-robust Generalised Method of Moments spatial error model from *spreg*, *GM_Error_Het*, based on Arraiz et al’s (2010) work on the remaining set of predictors.

After controlling for spatial autocorrelation, the percentages of minor roads, ‘B’ roads and primary roads as well as log-transformed median AADF becomes insignificant. The following are the summary of relationships between the remaining significant continuous predictors and median actual potential accessibility.

- a 10-percentage point (p.p.) increase in proportion of ‘A’ roads in an MSOA is associated with a potential accessibility score increase of about 11.4%
- a 1-p.p. increase in proportion of roads with bus lanes in an MSOA is associated with a potential accessibility score increase of about 8.4%
- increasing junction density in an MSOA by 10% is associated with a potential accessibility score increase of about 3.3%
- decreasing bus stop density in an MSOA by 10% is associated with a potential accessibility score increase of about 1.6%
- increasing median trips that serves an MSOA by 10% is associated with a potential accessibility score increase of about 1.9%
- every kilometre that an MSOA is nearer to the city centre is associated with a potential accessibility score increase of about 14.7%

The final model has a pseudo R-squared value of 0.7863.

Table 1: Regression Results on log-transformed Overall Actual Potential Accessibility

MODEL TYPE	OLS	OLS	OLS	GM Error Het
Adjusted/Pseudo R ²	0.796	0.797	0.796	0.786
Bristol (Cat Var)	0.0022 (0.979)			
Leeds (Cat Var)	0.0397 (0.592)			
Pct of Minor Roads	1.3548 (0.006)	1.3799 (0.004)	1.5270 (0.001)	-0.0089 (0.981)
Pct of B-Roads	2.1540 (0.003)	2.1321 (0.003)	2.2996 (0.001)	0.9488 (0.107)
Pct of A-Roads	1.6034 (0.015)	1.5601 (0.017)	1.8956 (0.002)	1.0805 (0.028)
Pct of Dual-Carriageway Roads	1.8254 (0.145)	1.8949 (0.128)		
Pct of Primary Roads	2.3606 (0.003)	2.3705 (0.003)	2.7465 (< 0.001)	0.8772 (0.205)
Pct of Roads with Bus Lanes	16.5309 (< 0.001)	16.6417 (< 0.001)	18.1395 (< 0.001)	8.0717 (0.010)
Junction Density (logged)	0.5238 (< 0.001)	0.5186 (< 0.001)	0.5168 (< 0.001)	0.3377 (< 0.001)
Bus Stop Density (logged)	-0.2660 (< 0.001)	-0.2639 (< 0.001)	-0.2527 (0.001)	-0.1465 (0.012)
Median AADF (logged)	-0.1644 (0.001)	-0.1685 (0.001)	-0.1425 (0.002)	-0.0691 (0.156)
Median Trips (logged)	0.1976 (< 0.001)	0.1971 (< 0.001)	0.2018 (< 0.001)	0.1940 (< 0.001)
Median Euclidean Distance	-0.1465 (< 0.001)	-0.1472 (< 0.001)	-0.1485 (< 0.001)	-0.1370 (< 0.001)
Lambda				0.1416 (< 0.001)

With reference to Table 2, the OLS regression on TTI rates found that the city categorical predictor, percentages of minor roads and primary roads in the MSOA, as well as the log-transformed median trips that serve the MSOA are significant at 10% level. OLS diagnostic testing found these set of predictors to be heteroskedastic (Koenker-Basset $p < 0.001$), spatial autocorrelation was found to be weakly significant (Moran's $I = 0.1864$, $p < 0.001$) and robust Lagrange Multiplier tests pointed towards spatial error (6.2, $p = 0.0128$, compared to robust spatial lag = 0.006, $p = 0.9388$).

Therefore, *GM_Error_Het* is also used on the final set of predictors.

After controlling for spatial autocorrelation, it can only be said that Bristol has a TTI rate that is about 5.6-percentage points lower than Manchester. Leeds is statistically indistinguishable from Manchester. Meanwhile, percentage of primary roads in the MSOA becomes insignificant. The following are the summary of relationships between the remaining significant continuous predictors and TTI rate.

- a 10-p.p. increase in proportion of minor roads in an MSOA is associated with a 0.63 p.p. reduction in TTI rates
- increasing median trips that serves an MSOA by 10% is associated with a 0.063 p.p. increase in TTI rates

The final model has a pseudo R-squared value of 0.2384.

Table 2: Regression Results on TTI Rates

MODEL TYPE	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	OLS	GM Error Het
Adjusted/Pseudo R ²	0.226	0.227	0.228	0.230	0.231	0.232	0.232	0.234	0.233	0.238
Bristol (Cat Var)	-0.0572 (< 0.001)	-0.0572 (< 0.001)	-0.0572 (< 0.001)	-0.0574 (< 0.001)	-0.0576 (< 0.001)	-0.0556 (< 0.001)	-0.0559 (< 0.001)	-0.0557 (< 0.001)	-0.0551 (< 0.001)	-0.0563 (< 0.001)
Leeds (Cat Var)	0.0085 (0.131)	0.0085 (0.125)	0.0085 (0.125)	0.0083 (0.131)	0.0080 (0.139)	0.0094 (0.063)	0.0096 (0.058)	0.0097 (0.056)	0.0097 (0.056)	0.0120 (0.177)
Pct of Minor Roads	-0.0607 (0.105)	-0.0606 (0.105)	-0.0595 (0.086)	-0.0609 (0.075)	-0.0623 (0.067)	-0.0609 (0.072)	-0.0535 (0.094)	-0.0567 (0.067)	-0.0649 (0.031)	-0.0631 (0.053)
Pct of B-Roads	-0.0044 (0.937)	-0.0044 (0.937)								
Pct of A-Roads	0.0535 (0.283)	0.0537 (0.281)	0.0548 (0.249)	0.0505 (0.258)	0.0478 (0.277)	0.0476 (0.278)	0.0532 (0.217)	0.0507 (0.235)		
Pct of Dual-Carriageway Roads	-0.0244 (0.797)	-0.0225 (0.803)	-0.0236 (0.791)							
Pct of Primary Roads	-0.1184 (0.050)	-0.1187 (0.049)	-0.1184 (0.049)	-0.1231 (0.032)	-0.1192 (0.034)	-0.1181 (0.036)	-0.1163 (0.038)	-0.1172 (0.036)	-0.0819 (0.085)	-0.0411 (0.472)
Pct of Roads with Bus Lanes	0.0159 (0.947)									
Junction Density (logged)	-0.0043 (0.441)	-0.0043 (0.439)	-0.0042 (0.439)	-0.0042 (0.445)	-0.0042 (0.443)	-0.0035 (0.511)				
Bus Stop Density (logged)	0.0046 (0.417)	0.0046 (0.415)	0.0045 (0.417)	0.0043 (0.432)	0.0043 (0.440)	0.0043 (0.434)	0.0013 (0.671)			
Median AADF (logged)	0.0016 (0.669)	0.0016 (0.670)	0.0016 (0.668)	0.0013 (0.720)						
Median Trips (logged)	0.0054 (0.108)	0.0054 (0.102)	0.0054 (0.101)	0.0053 (0.105)	0.0053 (0.100)	0.0064 (0.026)	0.0058 (0.033)	0.0064 (0.007)	0.0070 (0.003)	0.0066 (0.017)
Median Euclidean Distance	-0.0003 (0.508)	-0.0003 (0.509)	-0.0003 (0.504)	-0.003 (0.516)	-0.004 (0.486)					
Lambda										0.0845 (< 0.001)

With reference to Table 3, the OLS regression on TTV rates found that the study area categorical predictor, percentage of B-roads, primary roads and roads with bus lanes in the MSOA, log-transformed median AADF, as well as its median Euclidean distance to the city centre were significant at 10% level. OLS diagnostic testing found this set of predictors to be heteroskedastic (Koenker-Basset $p = 0.0027$), spatial autocorrelation was found to be moderately significant (Moran's $I = 0.3102$, $p < 0.001$) and robust Lagrange Multiplier tests pointed towards spatial error (29.952, $p < 0.001$, compared to robust spatial lag = 2.031, $p = 0.1541$). Therefore, *GM_Error_Het* is also used on the final set of predictors.

After controlling for spatial autocorrelation, it can only be said that Bristol has a TTV rate that is about 5.7-percentage points higher than Manchester. Leeds is statistically indistinguishable from Manchester. Additionally, the percentages of B-roads and primary roads in the MSOA become insignificant. The following are the summary of relationships between the remaining significant continuous predictors and TTV.

- a 10-p.p. increase in proportion of roads with bus lanes in the MSOA is associated with a 7.4 p.p. reduction in TTV rates
- decreasing average number of vehicles flowing through an MSOA per day by 10% is associated with a 0.1 p.p. reduction in TTV rates
- every kilometre that an MSOA is further from the city centre is associated with a 0.29 p.p. reduction in TTV rates

The final model has a pseudo R-squared value of 0.2085.

Table 3: Regression Results on TTV Rates

MODEL TYPE	OLS	OLS	OLS	OLS	OLS	OLS	OLS	GM Error Het
Adjusted/Pseudo R ²	0.207	0.208	0.209	0.210	0.208	0.207	0.205	0.209
Bristol (Cat Var)	0.0543 (< 0.001)	0.0528 (< 0.001)	0.05320 (< 0.001)	0.0530 (< 0.001)	0.0506 (< 0.001)	0.0514 (< 0.001)	0.0539 (< 0.001)	0.0567 (< 0.001)
Leeds (Cat Var)	0.0128 (0.073)	0.0123 (0.082)	0.0122 (0.084)	0.0116 (0.098)	0.0097 (0.160)	0.0112 (0.102)	0.0138 (0.036)	0.0119 (0.267)
Pct of Minor Roads	-0.0574 (0.230)	-0.0543 (0.251)	-0.0579 (0.215)	-0.0666 (0.145)				
Pct of B-Roads	-0.1600 (0.024)	-0.1561 (0.026)	-0.1602 (0.022)	-0.1710 (0.013)	-0.1379 (0.034)	-0.1260 (0.050)	-0.1263 (0.050)	-0.0813 (0.159)
Pct of A-Roads	-0.0877 (0.169)	-0.0817 (0.190)	-0.0834 (0.180)	-0.1041 (0.071)	-0.0751 (0.166)			
Pct of Dual-Carriageway Roads	-0.1054 (0.384)	-0.1007 (0.403)	-0.1064 (0.375)					
Pct of Primary Roads	0.1760 (0.023)	0.1778 (0.021)	0.1781 (0.021)	0.1565 (0.032)	0.1464 (0.044)	0.0931 (0.132)	0.1029 (0.094)	0.0790 (0.220)
Pct of Roads with Bus Lanes	-0.9752 (0.002)	-0.9537 (0.002)	-0.9561 (0.002)	-1.0414 (< 0.001)	-1.1010 (< 0.001)	-1.1724 (< 0.001)	-1.1835 (< 0.001)	-0.7378 (0.004)
Junction Density (logged)	-0.0048 (0.504)	0.0041 (0.559)	-0.0070 (0.082)	-0.0075 (0.057)	-0.0053 (0.148)	-0.0050 (0.169)		
Bus Stop Density (logged)	-0.0035 (0.627)	-0.0036 (0.617)						
Median AADF (logged)	0.0135 (0.006)	0.0136 (0.006)	0.0137 (0.005)	0.0122 (0.007)	0.0130 (0.004)	0.0139 (0.002)	0.0139 (0.002)	0.0097 (0.084)
Median Trips (logged)	0.0020 (0.639)							
Median Euclidean Distance	-0.0034 (< 0.001)	-0.0036 (< 0.001)	-0.0036 (< 0.001)	-0.0035 (< 0.001)	-0.0034 (< 0.001)	-0.0033 (< 0.001)	-0.0029 (< 0.001)	-0.0029 (< 0.001)
Lambda								0.0997 (< 0.001)

5.3: Travel Time Modelling on Hypothetical Scenarios

With reference to Figures 10-12 and Tables 4-6, regardless of study area, it would be better to pursue in-vehicle travel time reduction strategies instead of increasing bus frequencies because the mean improvement in potential accessibility to city centres from baseline conditions is higher for the former vis-à-vis the latter strategy (12.43% increase vs 10.67% increase in Bristol, 19.28% increase vs 11.93% increase in Leeds, and 9.81% increase vs 3.40% increase in Manchester).

Pertaining to the boost coming from encouraging multimodal journeys, Bristol could see a mean addition of only up to 2.85 p.p. in relative improvements to potential accessibility on top of whichever main strategy that they choose to adopt, while Leeds could see a higher mean addition of up to 5.03 p.p. On the other hand, Manchester's mean addition can go up to 11.66 p.p.

If only partial improvements could be made, Bristol should opt to prioritise city centre-bound patterns because the mean reduction is about 3.22 p.p. from whichever main strategy that they choose to adopt vis-à-vis a mean reduction of 4.68 p.p. if patterns serving most deprived MSOAs were prioritised. This situation flips for Leeds because the mean reduction from prioritising patterns serving most deprived MSOAs is only 1.91 p.p. in comparison to a mean reduction of 4.9 p.p. if city centre-bound services were prioritised. Manchester could choose either option because the average reduction from each partial strategy is quite minimal (0.96 p.p. for in-deprived MSOA services, 0.91 p.p. for city centre-bound services).

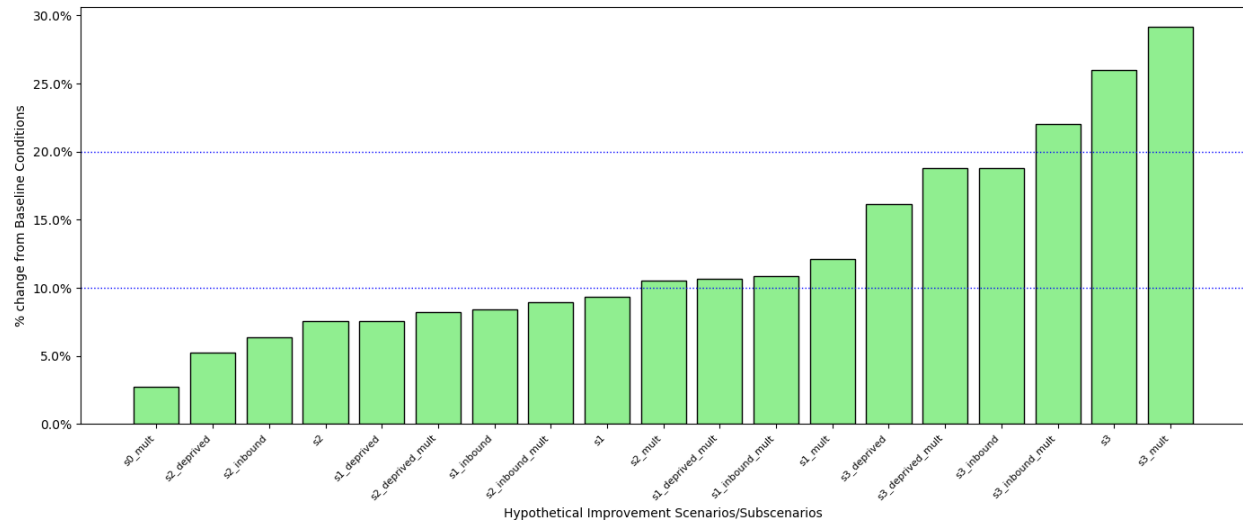


Figure 10: Bar Graph of Relative Improvements from Baseline Conditions based on Scenario/Sub-Scenario in Bristol
Data is also presented in table format in Appendix I

Table 4: Summary of Mean Relative Improvements by Scenario and Further Additions/Subtractions by Sub-Scenario in Bristol

Scenario	Mean Improvement (%)	Sub-Scenario	Mean Further Change (p.p.)
Reduce Travel Times	12.43	Encouraging Multimodal Journeys	+2.85
		In-Deprived MSOA Priority	-4.68
Increase Frequency	10.67	City Centre-Bound Priority	-3.22

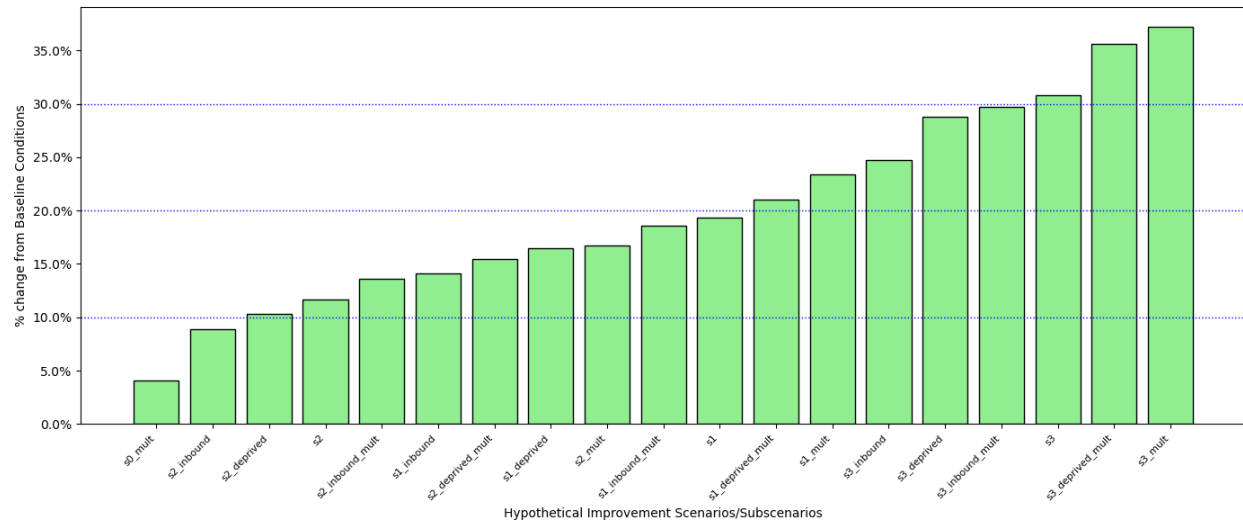


Figure 11: Bar Graph of Relative Improvements from Baseline Conditions based on Scenario/Sub-Scenario in Leeds
Data is also presented in table format in Appendix I

Table 5: Summary of Mean Relative Improvements by Scenario and Further Additions/Subtractions by Sub-Scenario in Leeds

Scenario	Mean Improvement (%)	Sub-Scenario	Mean Further Change (p.p.)
Reduce Travel Times	19.28	Encouraging Multimodal Journeys	+5.03
		In-Deprived MSOA Priority	-1.91
Increase Frequency	11.93	City Centre-Bound Priority	-4.90

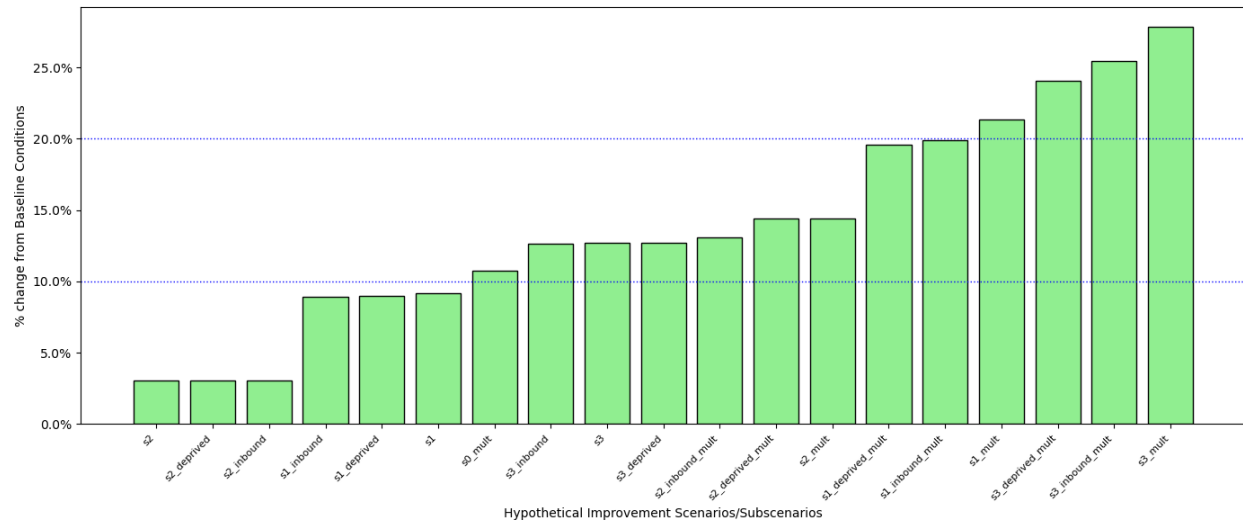


Figure 12: Bar Graph of Relative Improvements from Baseline Conditions based on Scenario/Sub-Scenario in Manchester
Data is also presented in table format in Appendix I

Table 6: Summary of Mean Relative Improvements by Scenario and Further Additions/Subtractions by Sub-Scenario in Manchester

Scenario	Mean Improvement (%)	Sub-Scenario	Mean Further Change (p.p.)
Reduce Travel Times	9.81	Encouraging Multimodal Journeys	+11.66
		In-Deprived MSOA Priority	-0.96
Increase Frequency	3.40	City Centre-Bound Priority	-0.91

5.4: Discussion on Possible Interventions

Regression results show that after accounting for locational factors (bus governance contexts and/or distance to city centre as well as spatial autocorrelation), road network characteristics generally have the greatest influence on various aspects of actual bus service performance. By having larger-capacity roads ('A' Roads), providing traffic with more opportunities to branch out via more junctions, and allowing buses to be separated from other traffic via bus lanes, it can facilitate faster in-vehicle travel times and therefore have a greater impact on an MSOA's overall actual potential accessibility to the city centre than having fewer bus stops or increasing trip counts. Additionally, that bus lane effect of separating buses from other traffic and thus improving bus service consistency – decreasing TTV rates – is also greater than just reducing median traffic flow in the area. Meanwhile, having a lot of smaller-capacity roads could slow down buses and make the TTI rates trend more negatively than reducing trip counts.

Comparing these set of results with current literature, especially research on factors influencing TTI and TTV via regression, is not directly possible due to differences in scope, dependent variables and factors evaluated. This paper looks at TTI and TTV as rates of actual potential accessibility across entire cities, while other papers look at TTI and/or TTV only in terms of the travel time itself and only for one or two specific routes. Therefore, while Kaewunruen, Sresakoolchai and Sun (2021) as well as Mazloumi, Currie and Rose (2010) have separately identified that, at route-level, having to make more stops at either bus stops or intersections slow the buses down and increases TTI and TTV respectively, this paper's regression, proxying for bus stops and intersection as densities per km² in each MSOA, found that neither have any statistically significant relationship with TTI or TTV at the MSOA level.

With regard to governance models, regression results also established that cities on EP model do not necessarily perform worse than the already-franchised Manchester – only Bristol has significantly lower TTI rates and higher TTV rates than Manchester, but Leeds is statistically indistinguishable on both aspects. Therefore, it seems to suggest that regardless of the bus governance context, an LTA wishing to improve potential accessibility, especially to the city centre, should generally focus on increasing its road capacity, specifically the capacity to accommodate dedicated bus flows away from, and over, other vehicular traffic. There would be more to gain from doing this than adopting a franchised model to directly modify bus networks and schedules. The hypothetical travel time modelling further supports this notion, with the results generally favouring

the outcome of transit priority infrastructure additions that reduce in-vehicle travel times over increasing frequencies for specific services.

However, all three study areas have been implementing transit priority infrastructure and improving their road networks in some form since the pandemic or before (Travelwest, 2014; West Yorkshire Combined Authority, 2021; Transport for Greater Manchester, 2025) since LTAs have the powers to carry out such interventions long before Bus Services Act 2025. Therefore, the overall actual median travel time (see Figure 6) and overall actual potential accessibility (see Figure 7) may already reflect some of these infrastructural additions. This also means that the mean relative improvements from Scenario 1 and its sub-scenarios derived from hypothetical travel time modelling is likely to be inflated because not all services can further reduce their in-vehicle travel time by 15 more percent when parts of their routes have already benefited from having bus lanes, guided busways or bus priority signalling at intersections.

What WECA and WYCA have not done was to intervene in the specific frequencies of their bus services. This would make the mean relative improvement figures for Scenario 2 and its sub-scenarios more robust, albeit still upper-bound counterfactuals. Meanwhile, regression results do point towards a significant and positive relationship between overall actual potential accessibility and the median number of trips serving an MSOA during morning peak hours, even though it is not as influential as road network characteristic predictors. On top of that, while having more trips does mean adding more vehicles on the roads, leading to higher median AADF counts which does have a significant and negative relationship with TTV rates, it is also plausible for there to be a net decrease in median AADF counts because commuters who would have been driving to the city centre may find travelling by bus to be more attractive due to the increased frequencies.

Hence, it strengthens the case for WECA and WYCA to consider bus franchising as their next move to increase potential accessibility to their city centres. Through franchising, WECA could then move to increase frequencies for city centre-bound patterns while WYCA could then increase frequencies for patterns serving deprived MSOAs. To that end, WYCA is already moving to franchise their buses starting 2027 to form the Weaver Network (West Yorkshire Combined Authority, no date).

It should be noted that the interventions for WECA and WYCA are proposed purely based on utilitarian principles, which focuses on maximising aggregate potential accessibility gain, instead of

egalitarianism which prioritises the most disadvantaged residents (Pereira, Schwanen and Banister, 2016). This is clearer in WECA where prioritising patterns serving the most deprived MSOAs would cost them 1.5 p.p. more in mean relative improvement than prioritising city centre-bound patterns (Table 4). It is in Leeds where valuations based on utilitarianism and egalitarianism happen to converge – prioritising patterns serving the most deprived MSOAs is both the more equitable and more efficient option between the two sub-scenarios (Table 5). Nevertheless, both authorities should note the limited effects of the proposed interventions when done in isolation, especially on its perceptibility by the commuters. With reference to Figure 13, the travel time savings that a typical MSOA in Bristol or Leeds could gain from these limited interventions are about two minutes and one minute respectively.

In contrast, Manchester's larger gap between Scenario 1 and 2 figures in Table 6 is a direct consequence of GMCA already taking control of their buses and increasing frequencies for some service patterns. It meant that as of June 2026, there are not many patterns in Manchester that are deemed 'moderate frequency' and qualify for hypothetical trip injections under Scenario 2. Hence, the mean relative improvements for Scenario 2 in Manchester is only about 4.16% better than baseline conditions. Instead, GMCA should consider encouraging multimodal journeys as their next move to increase potential accessibility. This is because, with reference to Figure 9 and Appendix I, the relative improvements from baseline conditions when travel time modelling is allowed to consider up to two public transport rides exceeds 10%. This is more than when travel time modelling considers Scenario 1, 2, or their sub-scenarios separately but with only one public transport ride. Encouraging multimodal journeys is also more likely to be perceptible by the typical commuter, who could benefit from about five minutes in median travel time savings (see Figure 13). To that end, GMCA has already introduced fare-capping between the Bee Network (buses) and Metrolink (trams) and will progressively extend it to rail by 2030 (Transport for Greater Manchester, no date). However, they could do more by further amending bus schedules to better synchronise with tram and rail schedules (or the other way around) to facilitate seamless inter-modal transfers for commuters.

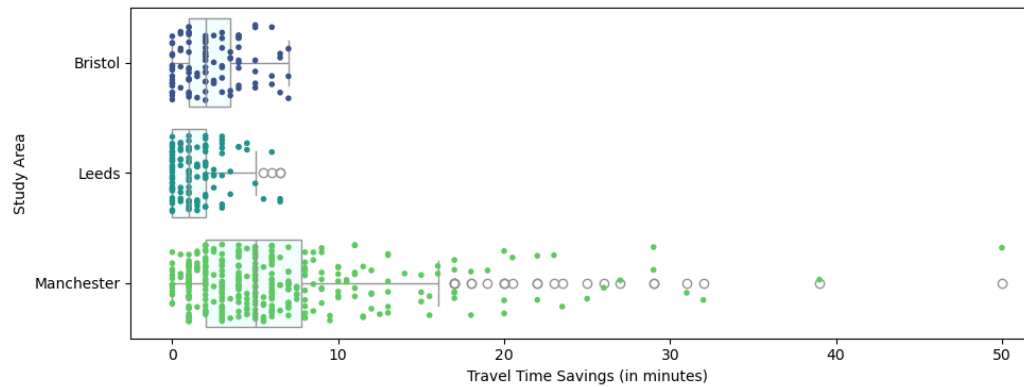
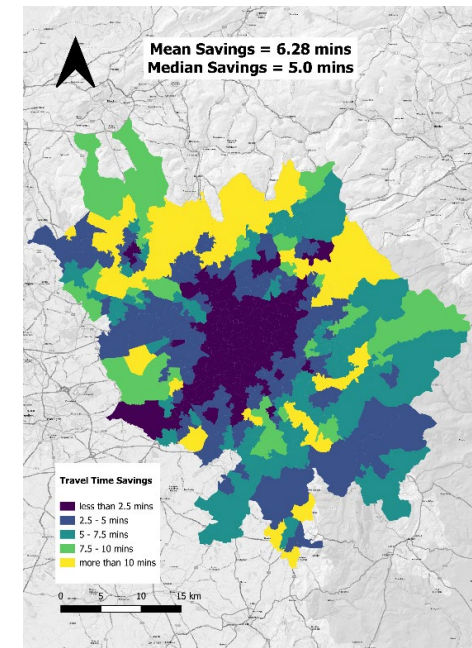
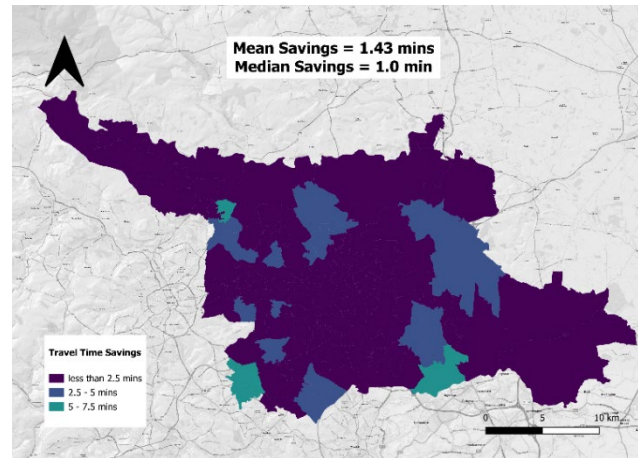
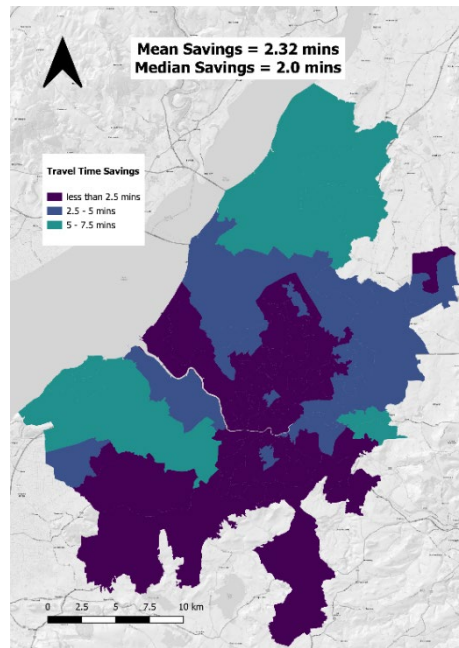


Figure 13: Travel Time Savings across Study Areas

Top-left – Savings from increasing frequency for city centre-bound patterns in Bristol

Top-centre – Savings from increasing frequency for patterns serving deprived MSOAs in Leeds

Top-right – Savings from encouraging multimodal journeys in Manchester

Bottom – Box and strip plot of travel time savings by study area

SECTION 6: LIMITATIONS AND CONCLUSION

6.1: Summary of Findings

This paper aims to explore factors that are associated with actual bus service performance across three English cities operating under different governance models and quantify the accessibility gains available from a set of hypothetical improvement scenarios, so as to speculate on the possible interventions that should be adopted in different study areas.

After controlling for spatial autocorrelation and inter-city differences, road network characteristics were the most influential predictors of potential accessibility, TTI and TTV, generally outweighing bus-network-specific characteristics (Section 5.2). Regarding the different bus governance models, the results also show that running an EP model does not necessarily lead to worse outcomes on bus service performance relative to the franchising model. This is because only Bristol's worse TTI and TTV rates relative to Manchester's are deemed statistically significant, while Leeds is statistically indistinguishable from Manchester.

The hypothetical travel time modelling also favoured transit-priority infrastructure investment over frequency increases in headline terms. However, modelled benefits from the former are likely to be inflated in corridors that have already benefited from bus priority measures, making frequency-based improvements the more robust, albeit modest, counterfactual (Section 5.3). Therefore, this points towards differentiated recommendations:

- While being on EP is not statistically worse than franchising, WECA and WYCA should still consider franchising to assume direct control in setting frequency – targeted at city centre-bound and deprived-MSOA-serving patterns respectively.
- GMCA, having already captured much of the readily available frequency-based gains through the Bee Network, would likely benefit more from multimodal fare and/or schedule integrations than either infrastructure-specific or bus-specific interventions.

6.2: Limitations and Directions for Future Research

The limitations of this paper that can inform future research directions can be classified into two broad categories – data source quality and methodology.

With regard to data source quality, there are two points that presented clear obstacles when conducting the analysis for this paper.

- Firstly, this paper defines each study area based on modified Census 2011 TTWA boundaries (Section 3) to reflect actual commuting patterns to the respective city centres. To date, this remains the most recent data and it is unclear if the Office for National Statistics would update it to reflect the data collected from Census 2021 or would not do so due to the fact that Census 2021 was collected during the tail-end of the third national lockdown, severely disrupting commuting patterns. On top of that, as seen in Figures 1-3, the TTWA boundaries also do not reflect the political boundaries of the combined authorities, which define the limits that LTAs can implement interventions. Therefore, future research may wish to explore alternative ways to define study areas that balances the boundaries where LTAs can exercise their powers but also reflects the updated post-pandemic commuting patterns to city centres.
- Secondly, every reported potential accessibility score, and every relative improvement percentage derived from it, depends on the distance-decay parameter β that was calibrated against Zhong et al's (2025) mobile-phone-derived MSOA-to-MSOA OD flow matrix (Section 4.3). While this paper considers the quality of the mobile-phone data considerably higher than Census 2021's Travel-To-Work data, it is still outdated and less likely to capture the full extent of commuting patterns that have evolved rapidly in the post-pandemic years. Future research with access to more recent OD data could recalibrate β and test how sensitive this paper's substantive conclusions are to that choice.

With regard to methodology, there are five points of differences that sets this paper apart from literature. While the exploratory nature of this paper in addressing a newly emerging opportunity and challenge for LTAs justifies its current methodological choices, future research would do well to explore these alternative methods to corroborate or challenge the findings of this paper:

- Firstly, the potential accessibility measure treats city-centre employment opportunities as homogeneous, weighted only by population, without regard to whether the type of job available matches the socio-economic profile of the residents being measured (Section 4.3). However, we know that Verduzco-Torres and McArthur (2026) show, in the context of

Greater Mexico City, that revising potential accessibility to account for this kind of employment matching produces materially different results from the standard unweighted measure, particularly for lower-educated populations. Therefore, applying an equivalent employment-matching weighting for Bristol, Leeds and Manchester could be another possible avenue for future research. For example, considering how jobs in the city centre are geared towards the new economy and are therefore more accessible for high-skilled workers (Section 1), would Leeds still benefit more by prioritising patterns serving deprived MSOAs than city centre-bound patterns?

- It should also be noted that the TTV rate definition adopted by this paper – the maximum range in actual potential accessibility scores across four different Wednesdays (Section 4.4) – make it highly sensitive to a single unusually good or bad Wednesday. Future research could consider adopting Braga, Loureiro and Pereira (2023) approach of collecting real-time data over more days and getting distributions of travel times between any two consecutive stops. This would facilitate the building of one single retrospective GTFS data based on the median travel times instead of multiple retrospective GTFS data per each day that is being studied. TTV could then be measured more robustly by building two more retrospective GTFS datasets using the travel times at, say, the 75th and 25th percentiles respectively.
- Thirdly, this paper adopts a stepwise backward when doing OLS regression on the various aspects of actual bus service performance (Section 4.5). This is a defensible starting point because backward elimination is a widely used method across different fields and, specifically for this paper, it facilitates the interpretation of relationships between predictors and bus service performance aspects. However, this method is also generally criticised because of the multiple hypothesis testing problem, where the final set of predictors are likely to have biased coefficient values, inflated significance in relation to the dependent variable, and could have been assembled by chance based on when other ‘non-significant’ predictors are eliminated in the earlier stages (Whittingham et al., 2006). Future research could instead apply LASSO or elastic net regularisation, which either performs variable selection intrinsically via a cross-validated penalty rather than a significance cutoff on each predictor regardless of its independence (Tibshirani, 1996) or handles correlated predictors more comprehensively (Zou and Hastie, 2005). Either alternative regression

methods could support screening a wider set of candidate predictors, such as additional built-environment, weather, surrounding land-uses, socio-demographic, or operator-level variables not considered here, without the same post-selection concerns from backward elimination.

- Furthermore, the hypothetical improvement scenarios modelled in Section 4.6 requires a huge *ceteris paribus* assumption, especially about demand. *r5py* estimates what travel times and accessibility would look like if a given service change were implemented exactly as specified, but they do not model travellers' potential responses such as mode shift from private to public transport, route reassignment, or crowding on newly popular services. Erhardt et al (2024) demonstrate one way to move beyond this, combining a data-driven travel-demand-replication tool with *MATSim*, an agent-based transport simulation framework, to test bus-service-reallocation strategies in two cities in the United States, and to observe how simulated individual travellers change behaviour in response. Applying a comparable agent-based approach in future research move the evaluation from the upper-bound, supply-side estimates produced here toward demand-responsive estimates of realised ridership and accessibility gains, including any congestion or crowding feedback that a static, schedule-based approach such as *r5py* cannot capture.
- Lastly, as indicated in Section 4.7 and demonstrated in Section 5.4, this paper evaluates hypothetical scenarios and sub-scenarios based on utilitarian principles. It does not consider the socio-geographical context where the benefits from the proposed interventions are likely to be received within a study area – whether it will benefit already well-connected MSOAs or a deprived one. While this paper attempts to incorporate elements of egalitarianism into hypothetical scenario by having the first sub-scenario (Section 4.6), the modelling results show that it only converges in the context of Leeds (Table 5). While incorporating employment matching into potential accessibility, as stated earlier, may address this, further research could also benefit from adopting a Rawlsian maximin criterion (Pereira, Schwanen and Banister, 2016) and rank interventions based on the effects on the least accessible and/or most deprived MSOAs on top of looking at the city-wide figures.

Additionally, the current conclusion that EP is not statistically different than franchising on bus service performance is only based on three study areas. Including more study areas, especially when more cities have switched from EP to franchising would allow for a more robust comparison between both governance models.

6.3: Concluding Remarks

This paper's two-fold contribution – exploring significant correlates of actual bus service performance across governance contexts, and quantifying the relative accessibility gains derived from different improvement scenarios – offers LTAs with some evidence base for a question that they will increasingly have to answer as a result of the increased devolved powers on local transport: whether to direct limited capital towards improving road infrastructure, or pursuing franchising powers to unlock greater flexibility in scheduling and frequency. The analysis suggests that infrastructure, specifically bus lane provision, is the more consistently influential lever across all three cities and both governance models, while franchising's key value lies in giving LTAs direct control over frequency and the ability to sync with other public transport modes once infrastructure gains are largely exhausted.

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*APPENDIX I – Tables of Relative Improvements from Baseline Conditions based
on Scenario/Sub-Scenario in Study Areas*

Table 7: Full Table of Relative Improvements from Baseline Conditions based on Scenario/Sub-Scenario in Bristol

Scenario	% Improvement from Baseline	Summary Calculations
Baseline with Multimodal (s0_mult)	2.71%	<u>Mean Scenario 1 Improvement</u> = [s1 + (s1_mult – s0_mult) + (s3 – s2)] / 3 12.43%
Full Scenario 1 (s1)	9.37%	
Full Scenario 1 with Multimodal (s1_mult)	12.12%	
Scenario 1 for Deprived MSOA (s1_deprived)	7.58%	
Scenario 1 for Deprived MSOA with Multimodal (s1_deprived_mult)	10.70%	<u>Mean Scenario 2 Improvement</u> = [s2 + (s2_mult – s0_mult) + (s3 – s1)] / 3 10.67%
Scenario 1 for Inbound Patterns (s1_inbound)	8.41%	
Scenario 1 for Inbound Patterns with Multimodal (s1_inbound_mult)	10.85%	
Full Scenario 2 (s2)	7.53%	
Full Scenario 2 with Multimodal (s2_mult)	10.53%	<u>Mean Multimodal Improvement</u> = [s0_mult + (all scenarios_mult – all scenarios)] / 10 2.85%
Scenario 2 for Deprived MSOA (s2_deprived)	5.26%	
Scenario 2 for Deprived MSOA with Multimodal (s2_deprived_mult)	8.22%	
Scenario 2 for Inbound Patterns (s2_inbound)	6.40%	
Scenario 2 for Inbound Patterns with Multimodal (s2_inbound_mult)	8.93%	<u>Mean Deprived MSOA-Priority Reduction</u> = [(all scenarios_deprived – all full scenarios)] / 6 -4.68%
Full Scenario 3 (s3)	26.02%	
Full Scenario 3 with Multimodal (s3_mult)	29.17%	
Scenario 3 for Deprived MSOA (s3_deprived)	16.13%	
Scenario 3 for Deprived MSOA with Multimodal (s3_deprived_mult)	18.81%	<u>Mean City Centre-Priority Reduction</u> = [(all scenarios_inbound – all full scenarios)] / 6 -3.22%
Scenario 3 for Inbound Patterns (s3_inbound)	18.83%	
Scenario 3 for Inbound Patterns with Multimodal (s3_inbound_mult)	22.02%	

Table 8: Full Table of Relative Improvements from Baseline Conditions based on Scenario/Sub-Scenario in Leeds

Scenario	% Improvement from Baseline	Summary Calculations
Baseline with Multimodal (s0_mult)	4.05%	<u>Mean Scenario 1 Improvement</u> $= [s1 + (s1_mult - s0_mult) + (s3 - s2)] / 3$ 19.28%
Full Scenario 1 (s1)	19.35%	
Full Scenario 1 with Multimodal (s1_mult)	23.42%	
Scenario 1 for Deprived MSOA (s1_deprived)	16.51%	
Scenario 1 for Deprived MSOA with Multimodal (s1_deprived_mult)	21.04%	<u>Mean Scenario 2 Improvement</u> $= [s2 + (s2_mult - s0_mult) + (s3 - s1)] / 3$ 11.63%
Scenario 1 for Inbound Patterns (s1_inbound)	14.11%	
Scenario 1 for Inbound Patterns with Multimodal (s1_inbound_mult)	18.60%	
Full Scenario 2 (s2)	11.66%	
Full Scenario 2 with Multimodal (s2_mult)	16.75%	<u>Mean Multimodal Improvement</u> $= [s0_mult + (all\ scenarios_mult - all\ scenarios)] / 10$ 5.03%
Scenario 2 for Deprived MSOA (s2_deprived)	10.28%	
Scenario 2 for Deprived MSOA with Multimodal (s2_deprived_mult)	15.48%	
Scenario 2 for Inbound Patterns (s2_inbound)	8.91%	
Scenario 2 for Inbound Patterns with Multimodal (s2_inbound_mult)	13.63%	<u>Mean Deprived MSOA-Priority Reduction</u> $= [(all\ scenarios_deprived - all\ full\ scenarios)] / 6$ -1.91%
Full Scenario 3 (s3)	30.78%	
Full Scenario 3 with Multimodal (s3_mult)	37.21%	
Scenario 3 for Deprived MSOA (s3_deprived)	28.81%	
Scenario 3 for Deprived MSOA with Multimodal (s3_deprived_mult)	35.57%	<u>Mean City Centre-Priority Reduction</u> $= [(all\ scenarios_inbound - all\ full\ scenarios)] / 6$ -4.90%
Scenario 3 for Inbound Patterns (s3_inbound)	24.77%	
Scenario 3 for Inbound Patterns with Multimodal (s3_inbound_mult)	29.72%	

Table 9: Full Table of Relative Improvements from Baseline Conditions based on Scenario/Sub-Scenario in Manchester

Scenario	% Improvement from Baseline	Summary Calculations
Baseline with Multimodal (s0_mult)	10.75%	<u>Mean Scenario 1 Improvement</u> $= [s1 + (s1_mult - s0_mult) + (s3 - s2)] / 3$ 9.81%
Full Scenario 1 (s1)	9.18%	
Full Scenario 1 with Multimodal (s1_mult)	21.38%	
Scenario 1 for Deprived MSOA (s1_deprived)	8.96%	
Scenario 1 for Deprived MSOA with Multimodal (s1_deprived_mult)	19.60%	<u>Mean Scenario 2 Improvement</u> $= [s2 + (s2_mult - s0_mult) + (s3 - s1)] / 3$ 3.40%
Scenario 1 for Inbound Patterns (s1_inbound)	8.95%	
Scenario 1 for Inbound Patterns with Multimodal (s1_inbound_mult)	19.87%	
Full Scenario 2 (s2)	3.06%	
Full Scenario 2 with Multimodal (s2_mult)	14.38%	<u>Mean Multimodal Improvement</u> $= [s0_mult + (all\ scenarios_mult - all\ scenarios)] / 10$ 11.66%
Scenario 2 for Deprived MSOA (s2_deprived)	3.06%	
Scenario 2 for Deprived MSOA with Multimodal (s2_deprived_mult)	14.38%	
Scenario 2 for Inbound Patterns (s2_inbound)	3.06%	
Scenario 2 for Inbound Patterns with Multimodal (s2_inbound_mult)	13.08%	<u>Mean Deprived MSOA-Priority Reduction</u> $= [(all\ scenarios_deprived - all\ full\ scenarios)] / 6$ -0.96%
Full Scenario 3 (s3)	12.68%	
Full Scenario 3 with Multimodal (s3_mult)	27.84%	
Scenario 3 for Deprived MSOA (s3_deprived)	12.68%	
Scenario 3 for Deprived MSOA with Multimodal (s3_deprived_mult)	24.09%	<u>Mean City Centre-Priority Reduction</u> $= [(all\ scenarios_inbound - all\ full\ scenarios)] / 6$ -0.91%
Scenario 3 for Inbound Patterns (s3_inbound)	12.66%	
Scenario 3 for Inbound Patterns with Multimodal (s3_inbound_mult)	25.48%	